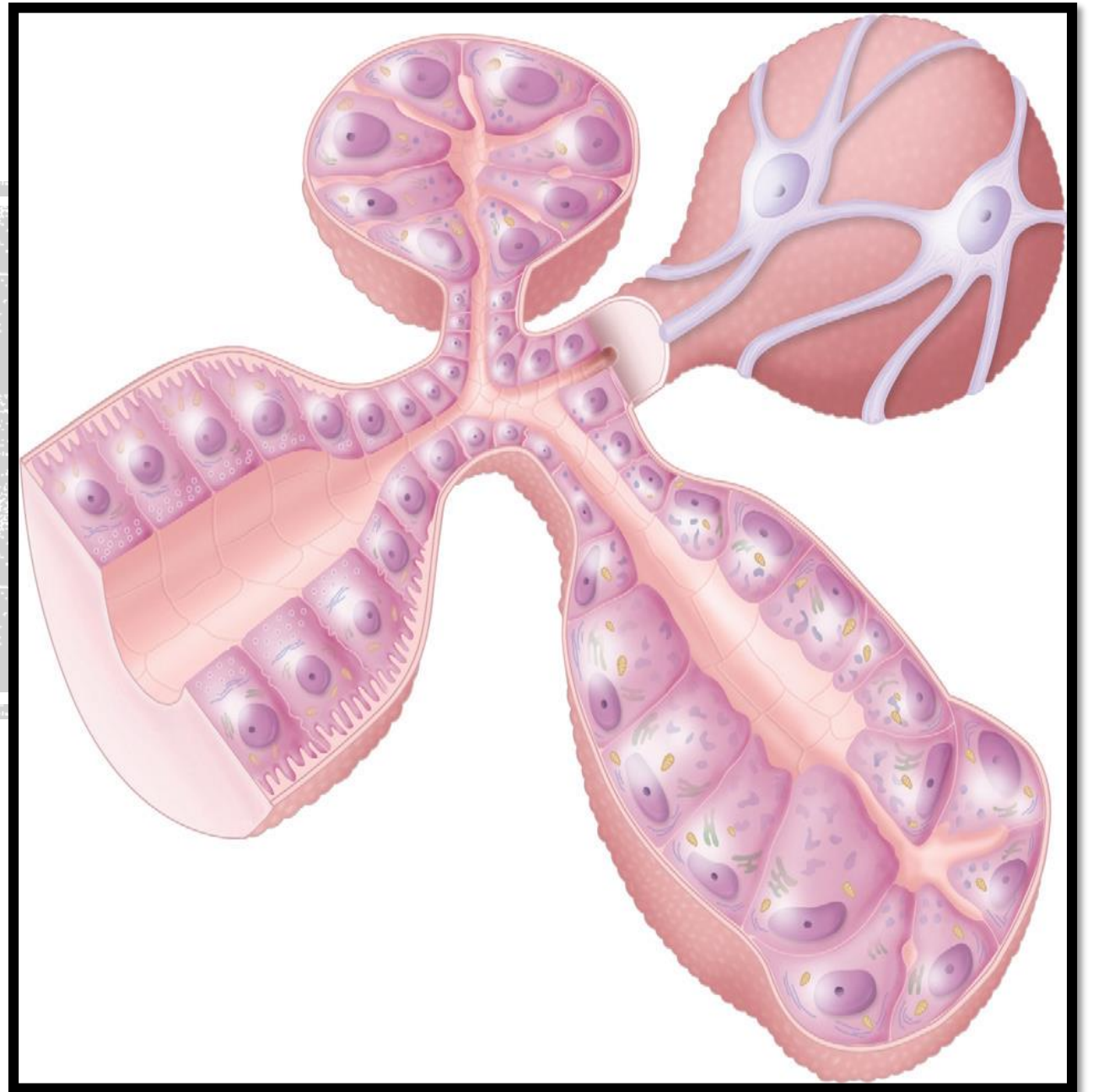


SALIVARY GLANDS

(HISTOLOGY)

Dr. Misbah Ashraf Moten



Learning Objectives:

Understand the histology of the salivary glands in terms of the parenchymal cells (mucous and serous) and

Understand the histology of the salivary glands in terms of the nature of the duct System

Be able to appreciate the differences between the three pairs of pairs of major salivary glands

Be able to describe the structure and possible function of the myoepithelial cells

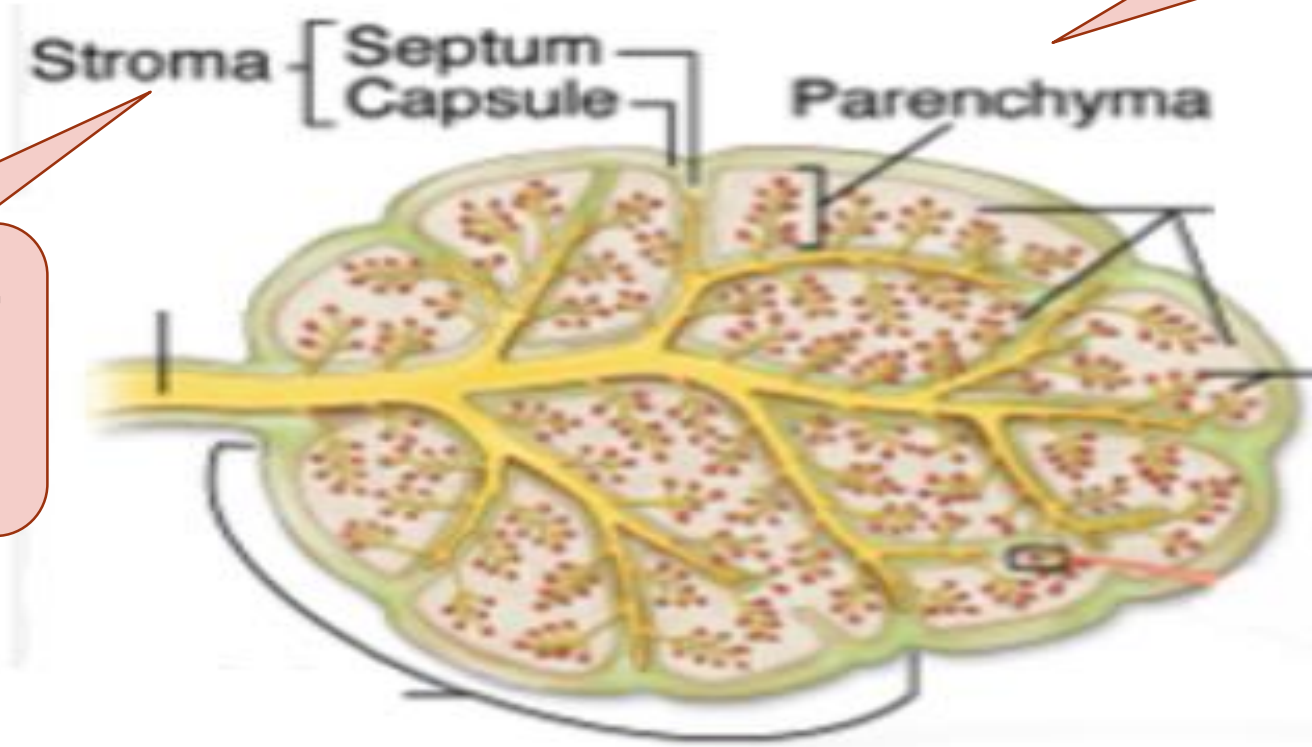
Relationship of structure & function of salivary glands to clinical correlations such as xerostomia



ELEMENTS OF SALIVARY GLANDS:

Glandular secretory tissue

**Supporting
connective
tissue**



GROSS MORPHOLOGY OF SALIVARY GLANDS:

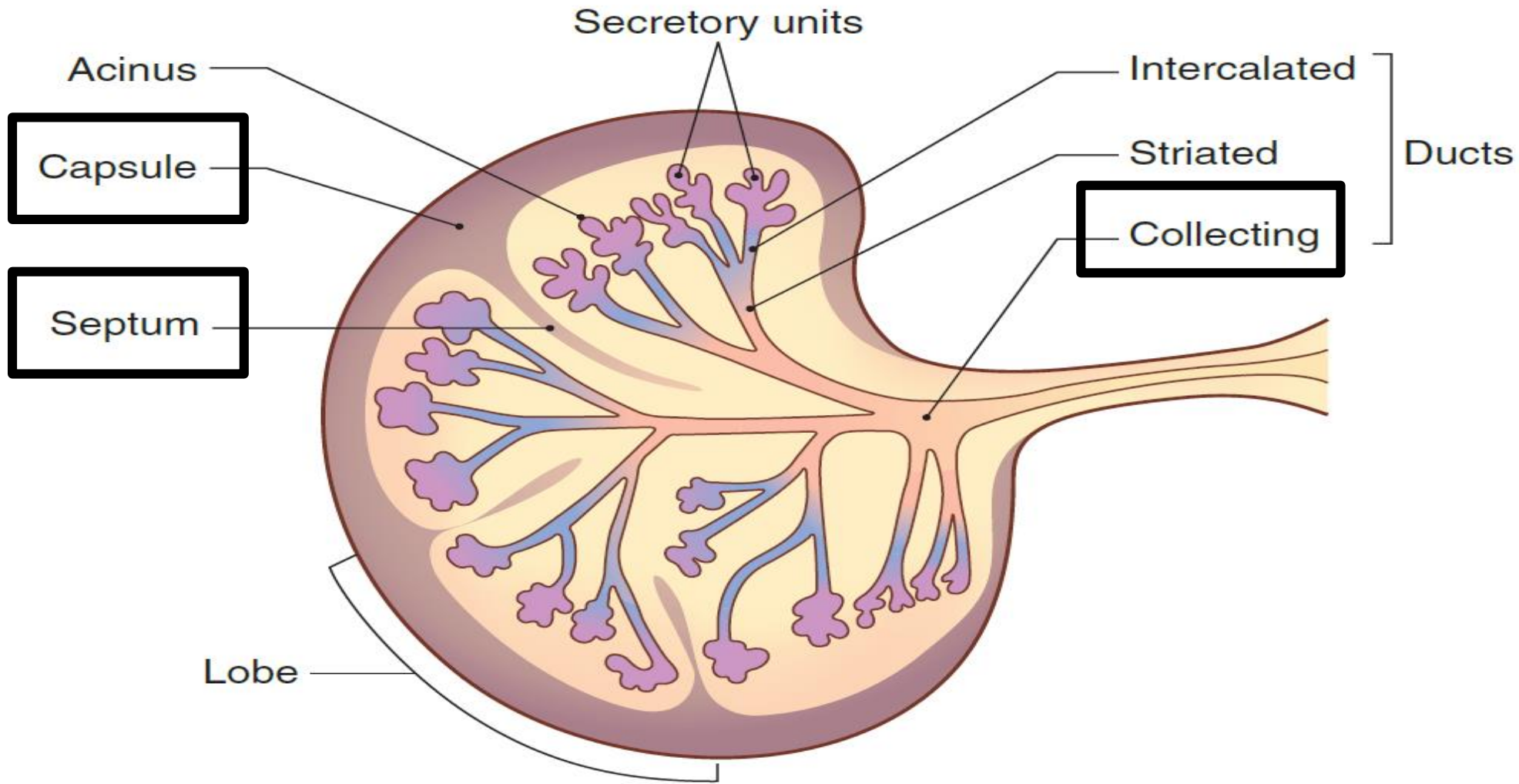
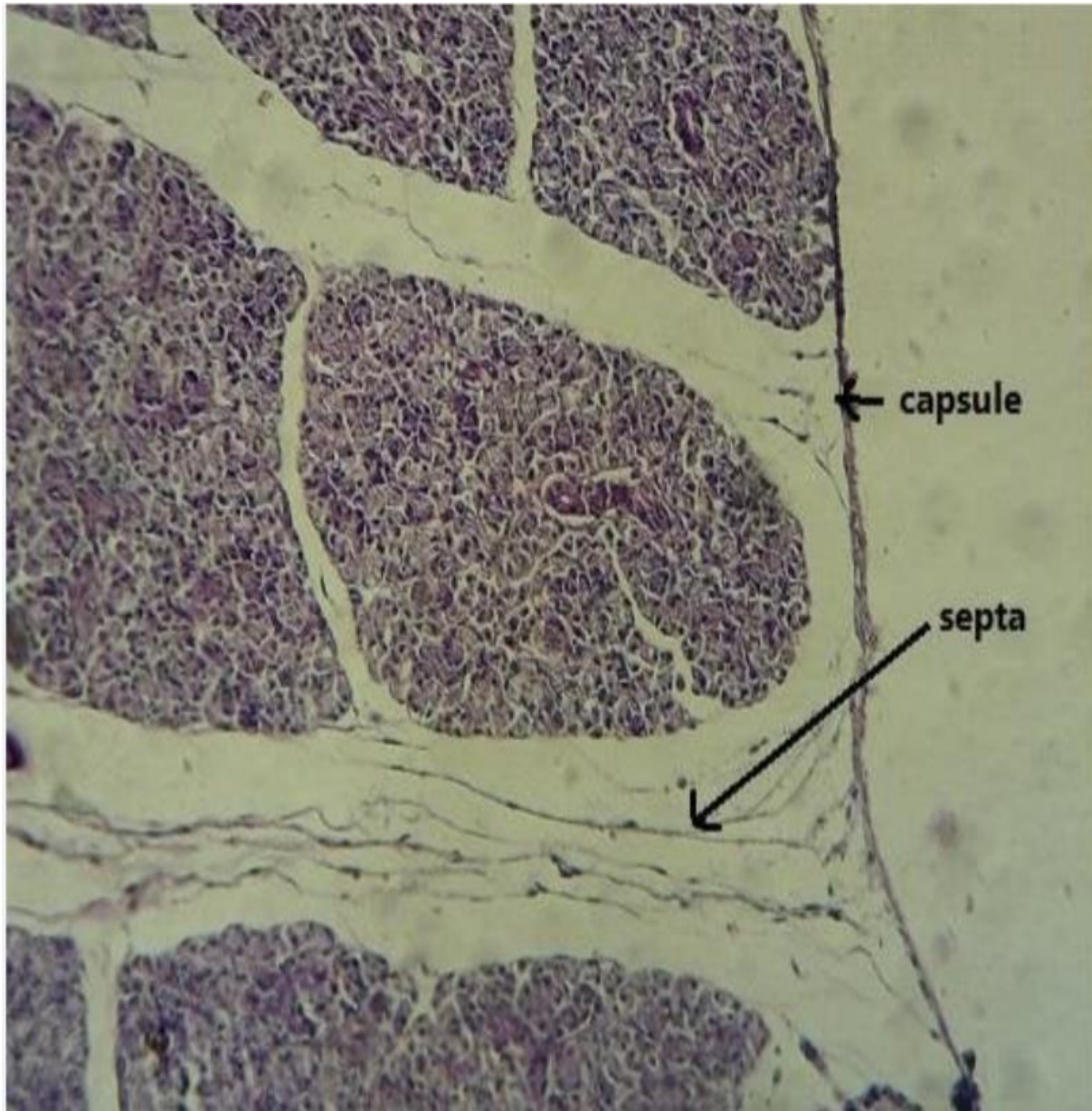


Fig. 16.1 The general organisation of a salivary gland.





Connective Tissue Septa contains:

- Fibroblasts,
- Collagen
- fat cells (much variable in parotid gland).
- Plasma cells (secrete immunoglobulins) - in the stroma of the gland around intralobular ducts.



GROSS MORPHOLOGY OF SALIVARY GLANDS:

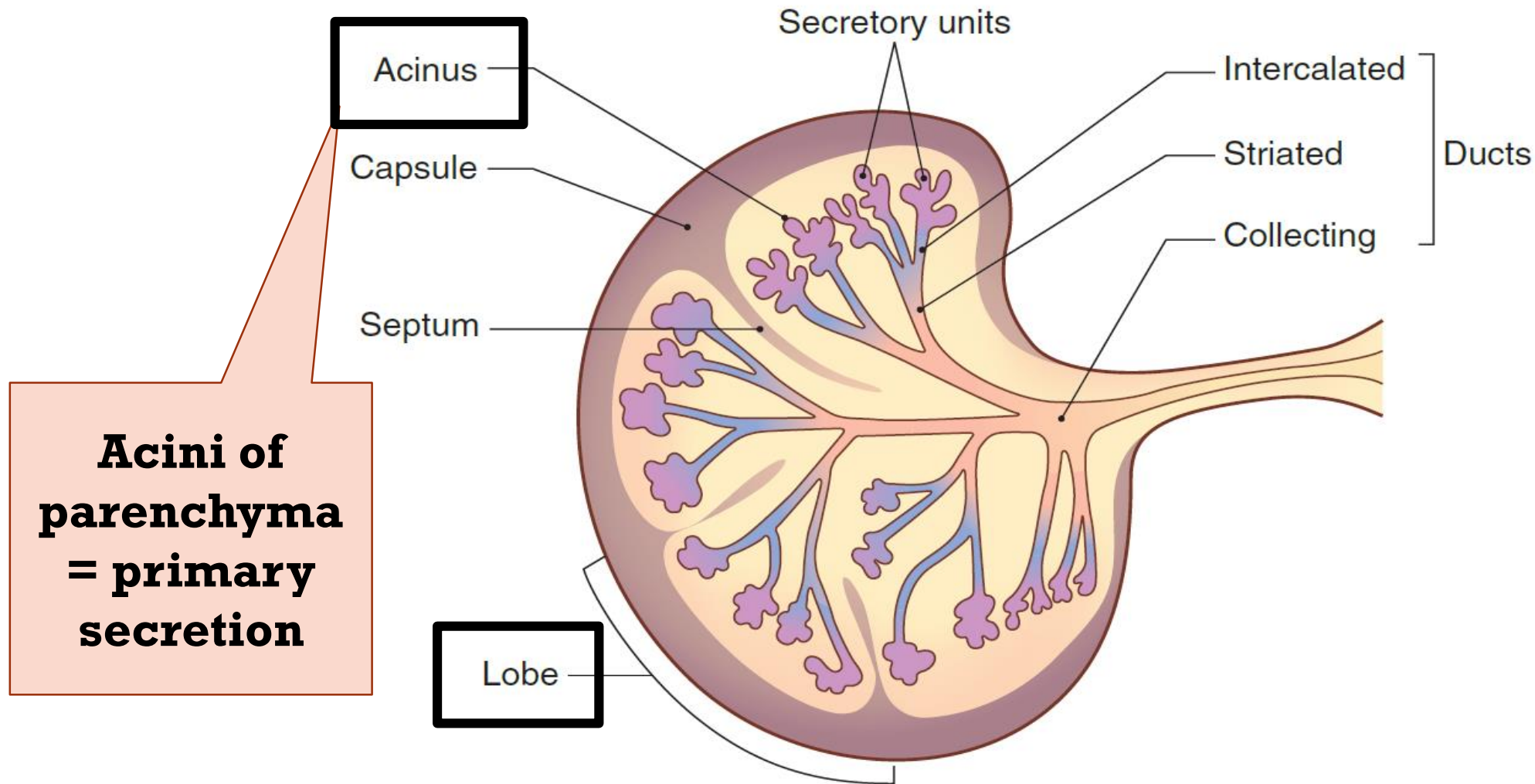
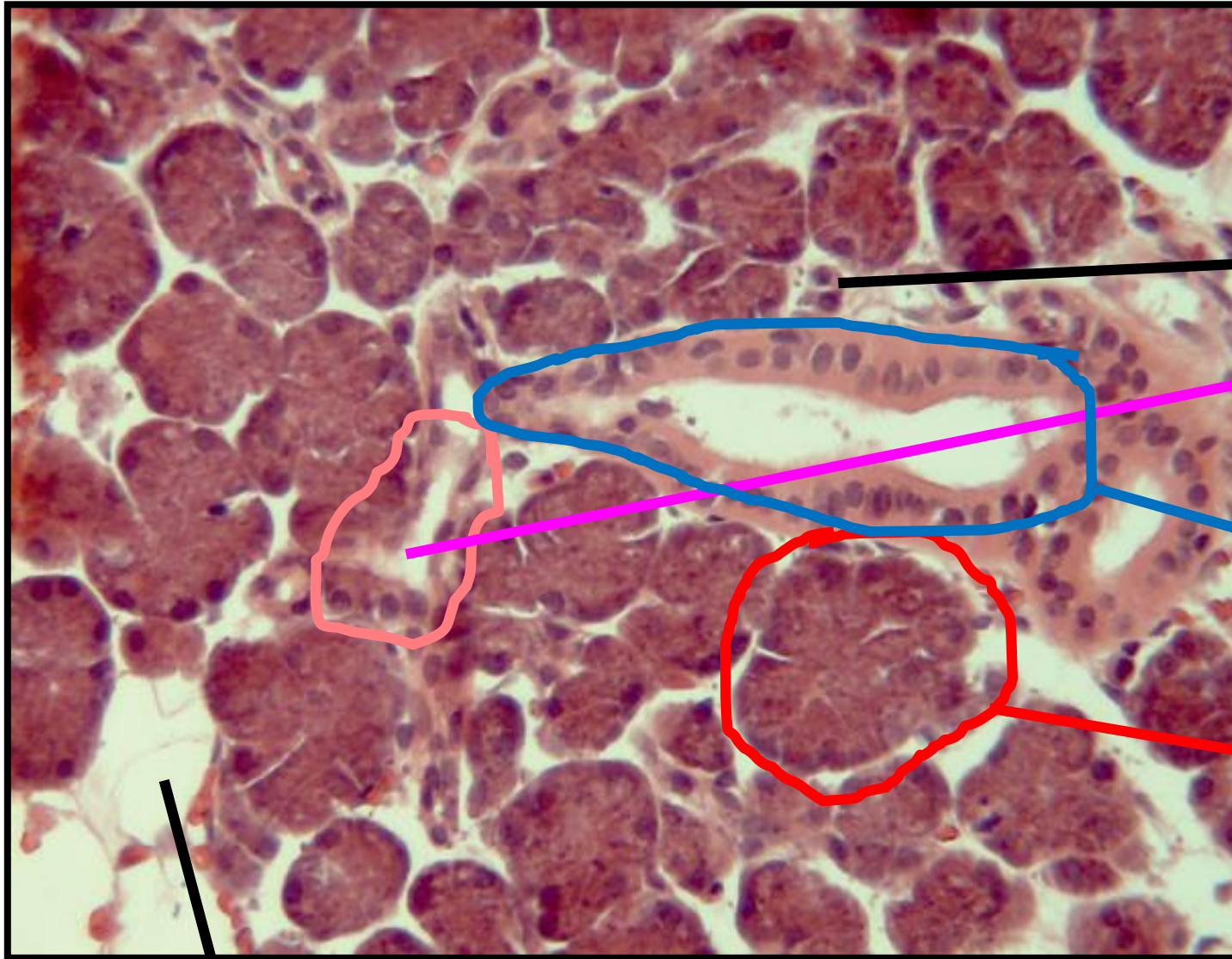


Fig. 16.1 The general organisation of a salivary gland.



SEROUS ACINI:

- produce a watery proteinaceous fluid
- source of amylase.



Plasma cell

Intercalated duct

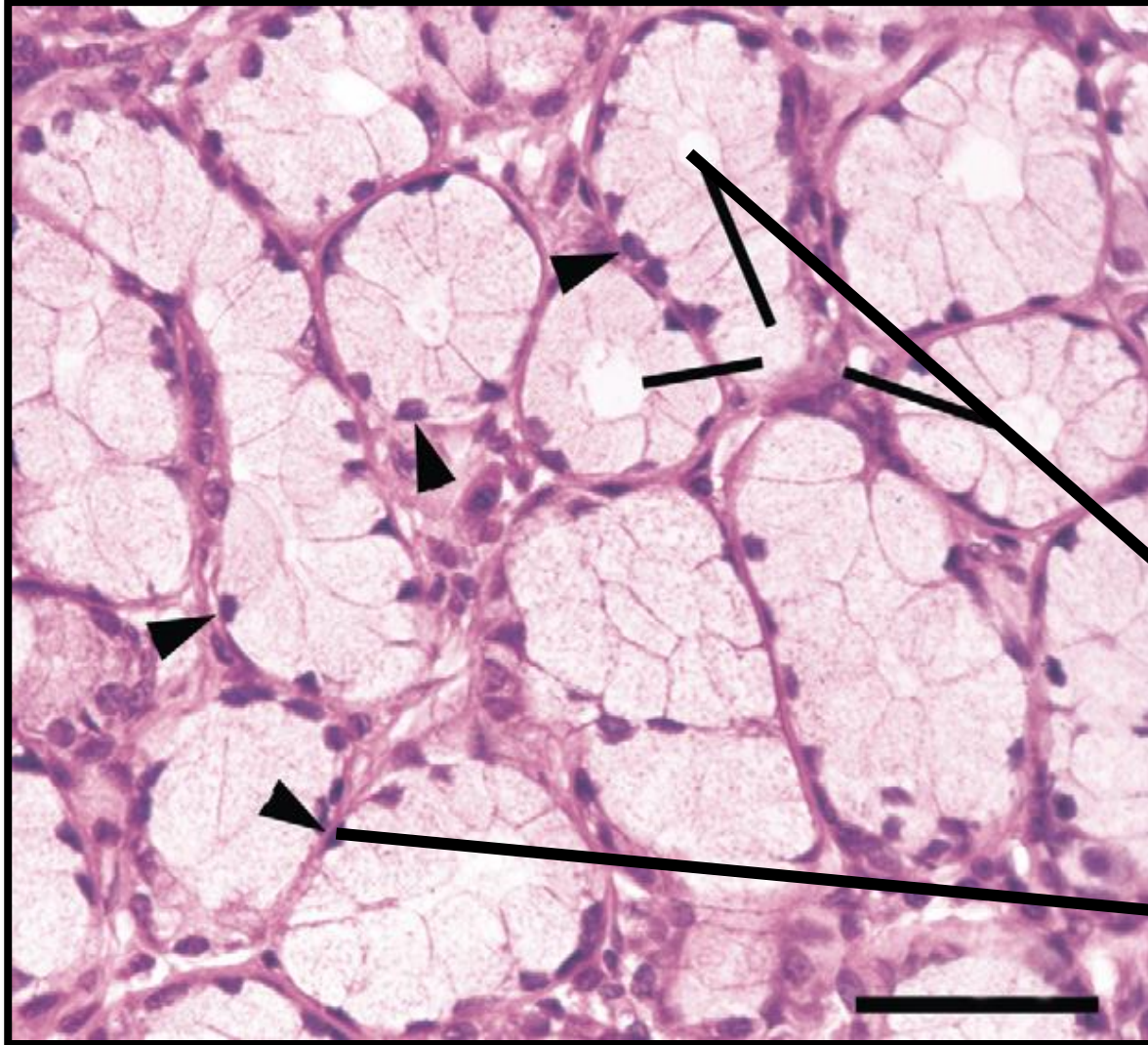
Striated Duct

Serous acini

Fat cells



MUCOUS ACINI:



- Contain proteins linked to $\uparrow$ carbohydrate content, forming a more viscous, mucin-rich product.
- More tubular form
- Poorly stained mucous secretory granules fill the cytoplasm
- Nuclei (flattened & compressed against basal surfaces of cells)
- Large lumina compared with those of serous acini.

Lumen of mucous cells

Nuclei

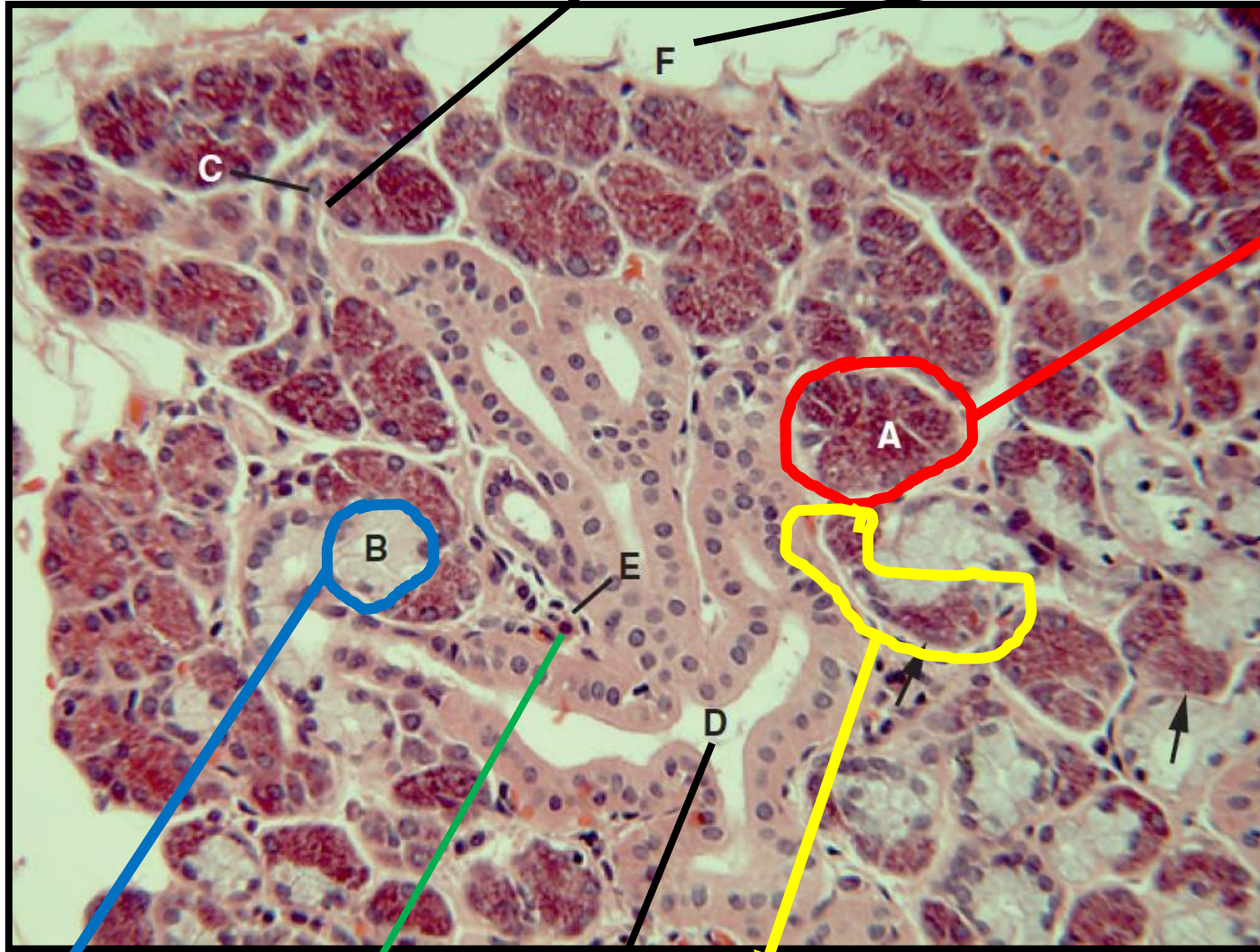


MIXED ACINI:

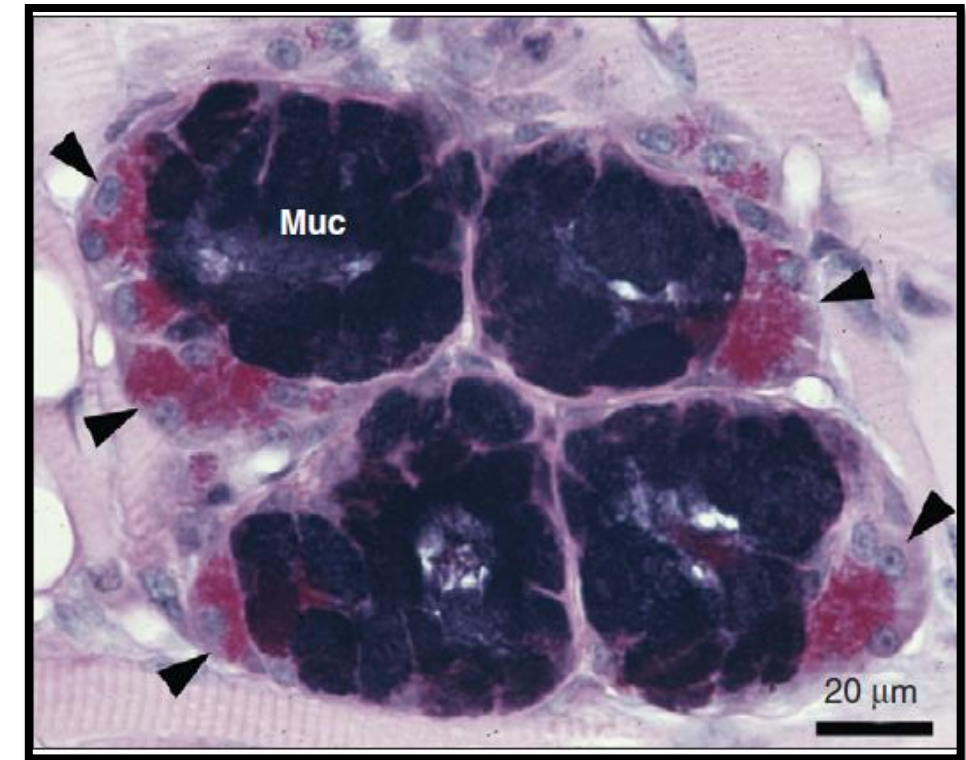
Intercalated
duct

Fat cells

serous cells form a cap
or demilune, outside the
mucous cells



Serous acini



Mucous acini

Plasma Cell

Striated duct

Serous Demilune

BASAL CELLS:

Location

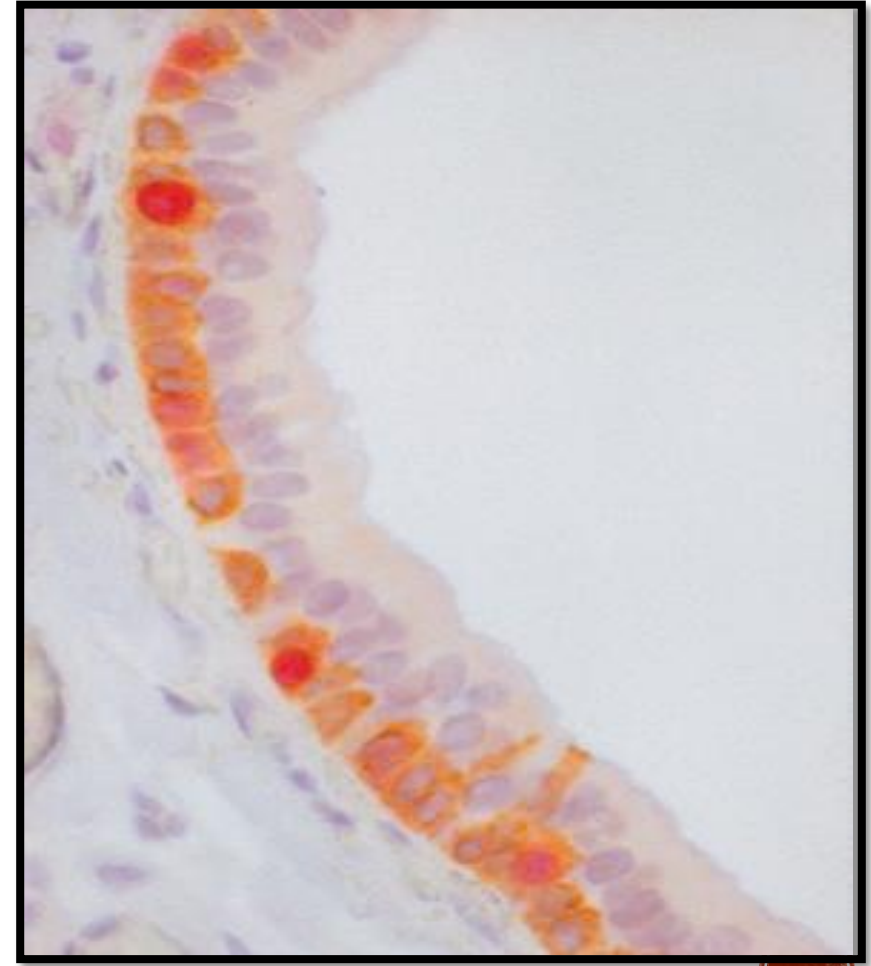
Present in **striated ducts** (sparse) & **collecting ducts** (more numerous) – forming continuous layer as ducts pass further towards hilum)

Distinguishing features from parenchymal cells:

- Morphology
- Co-expression of cytokeratin 14 and the antiapoptotic factor Bcl-2
- Highest proliferative index of any region = 3%
potential

Functions:

- stem cells during turnover,
- cell regeneration in salivary glands, during metaplasia (when oncocytes and sebaceous cells may appear)
- salivary gland regeneration (all parenchymal cells & myoepithelial cells) may also be involved



LYMPH NODES

- Lymph nodes are situated both on the surface and within the parotid gland but are not found within the other salivary glands.



MYOEPITHELIAL (BASKET) CELLS:

- Contractile cells with several processes, around the acini & intercalated ductal cells

Location: b/w basal lamina & basal membrane of acinar secretory cells & intercalated duct cells

Features:

- flattened nucleus
- Intracellular protein synthetic organelles → not particularly abundant.
- Absent Alkaline phosphatase in humans
- Myoepithelial cells (& basal cells of double-layered ducts) also contain cytokeratin (cytokeratin 17), intermediate filament & contractile actin filaments.

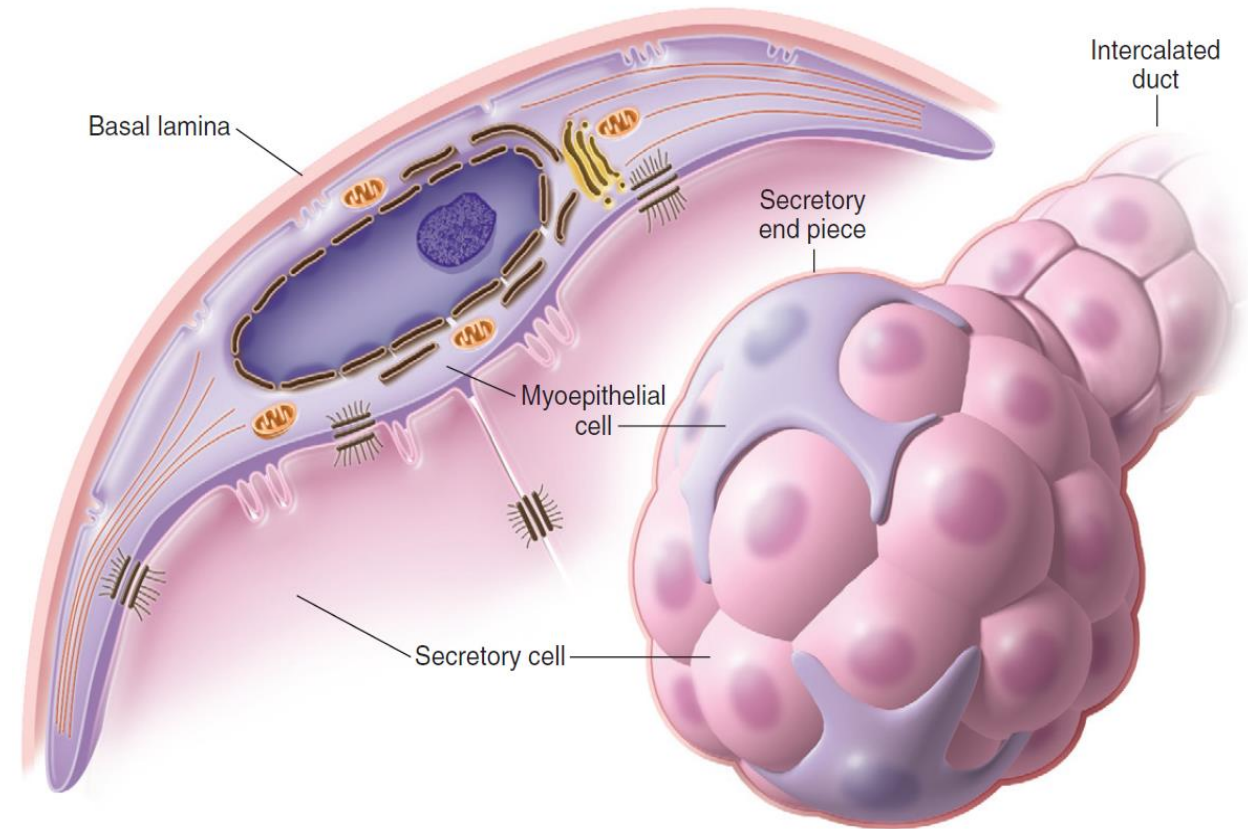


FIGURE 11-19 Myoepithelial cells in section (*top*) and surface (*bottom*) views.

Shape around acini

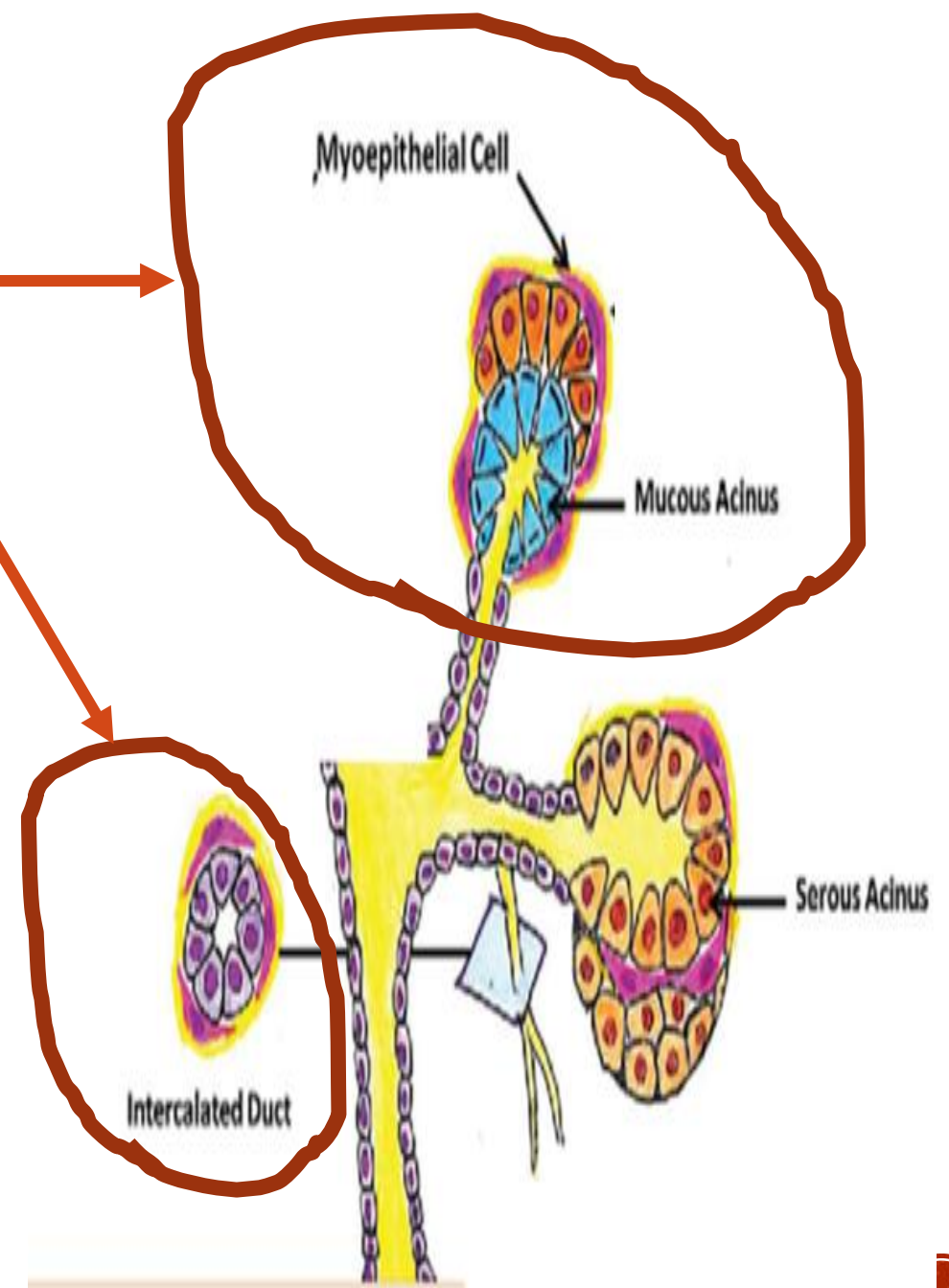
Dendritic cells, stellate shaped body

Some tapering processes lying in gutters on secretory cell surface, so acinar outline → smooth

Shape around intercalated ducts

Elongated cells, lying more superficially & running longitudinally along the duct

Produce a bulge in the outline of the duct & have few short processes



Box 16.2 **Myoepithelial activity within salivary glands**

It can:

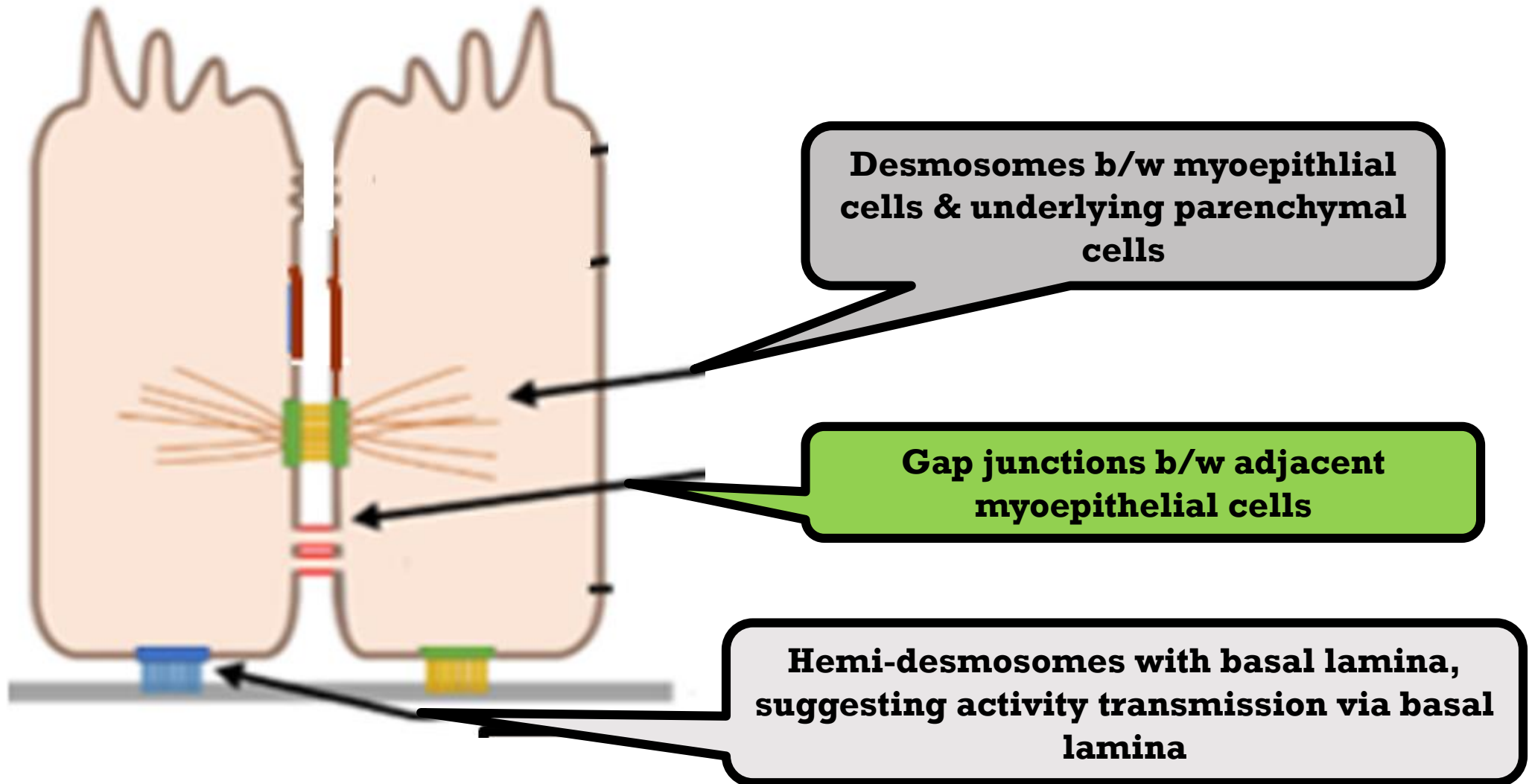
- support the underlying parenchyma and reduce back-permeation of fluid
- accelerate the initial outflow of saliva
- reduce luminal volume
- contribute to the secretory pressure
- help salivary flow to overcome increases in peripheral resistance – but if this is excessive it may lead to sialectatic damage of striated ducts, thereby increasing overall permeability.

CD44 expression on contacting surfaces of Myoepithelial cells with serous acinar cells (both cells), may play a role in basic processes associated with cell proliferation and differentiation, and thus function in normal & neoplastic state.

Assistance for some parenchymal cells to expel their contents and a milking effect on any underlying extracellular fluid, assisting passage via parenchymal tight junctions.



Pinocytotic vesicles and dense attachment areas (at parts of plasma membrane covered with basal lamina)



GROSS MORPHOLOGY OF SALIVARY GLANDS CONT.:

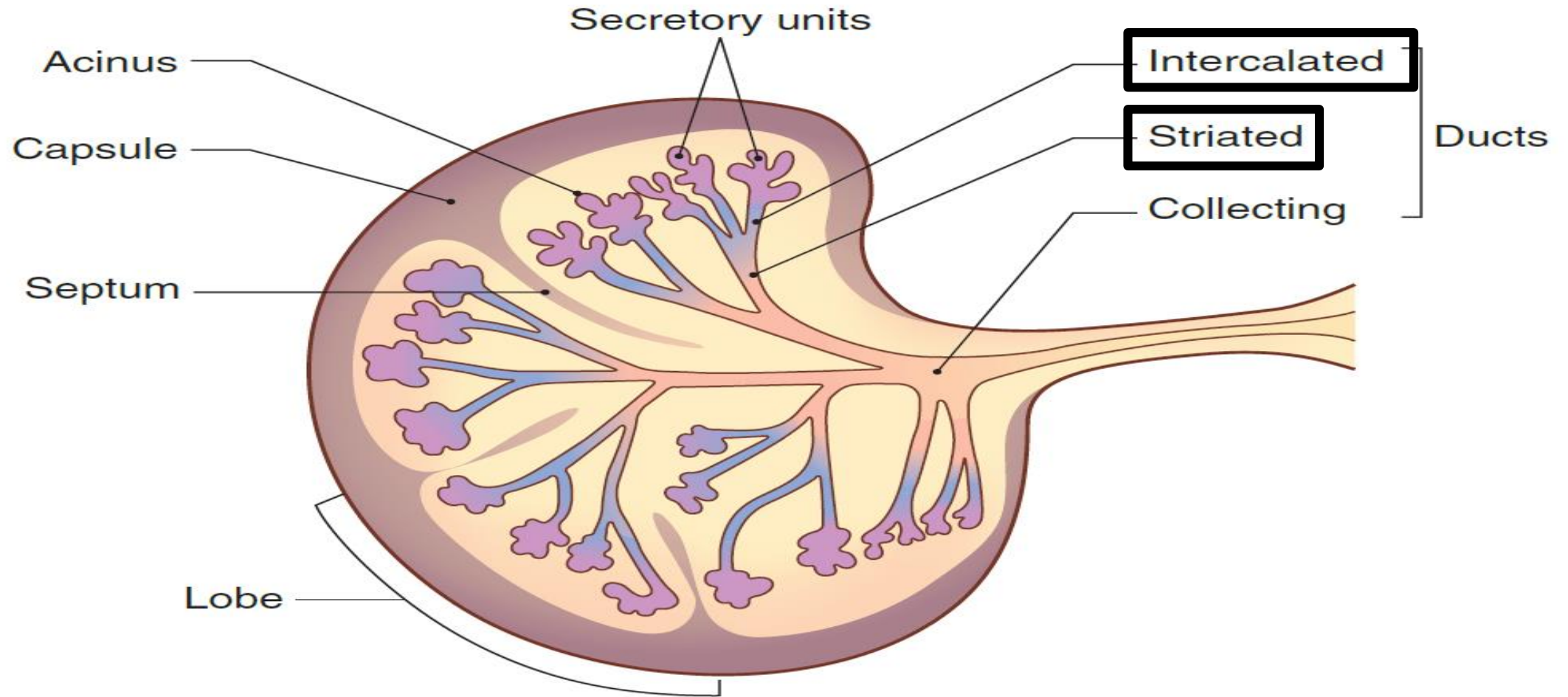
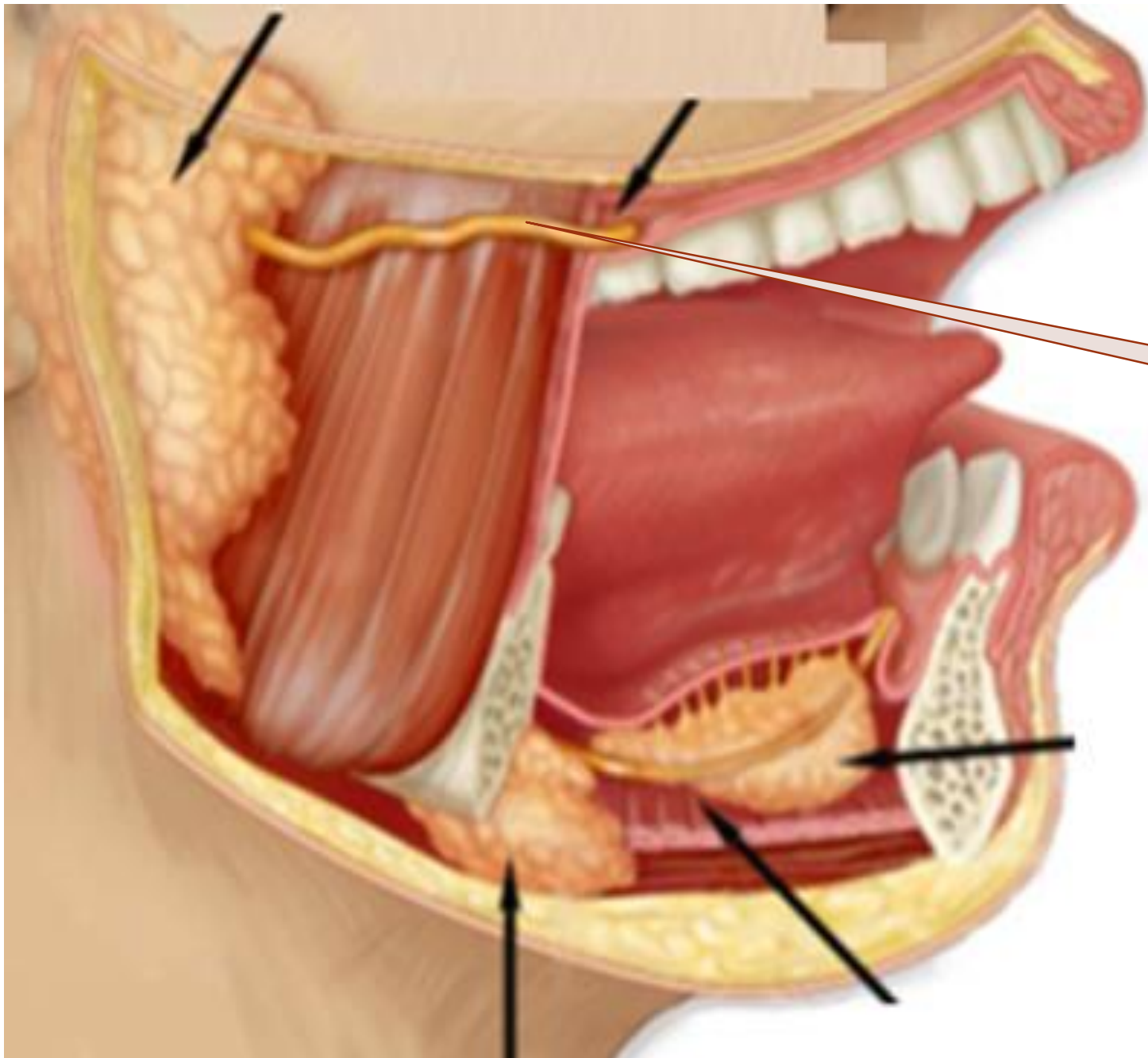


Fig. 16.1 The general organisation of a salivary gland.





- Collecting ducts



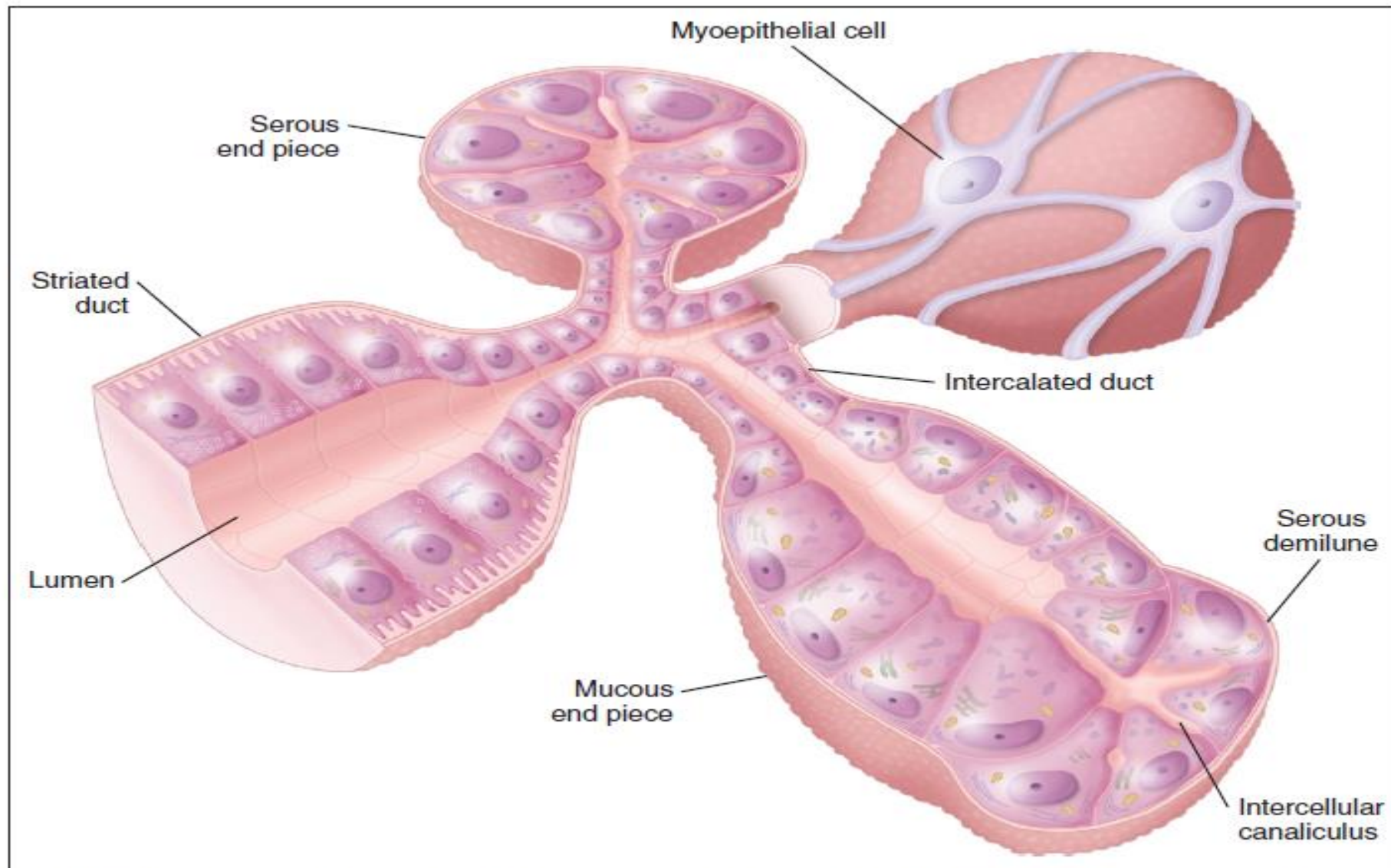


FIGURE 11-33 Architecture of salivary gland ducts and secretory end pieces and the main features of the parenchymal cells. (From Hand AR. In Bhaskar SN, editor: *Orban's oral histology and embryology*, ed 11, St Louis, 1991, Mosby.)



HISTOLOGY OF SECRETORY & DUCTAL ELEMENTS

contractile cells with several processes around the acini & intercalated ductal cells.

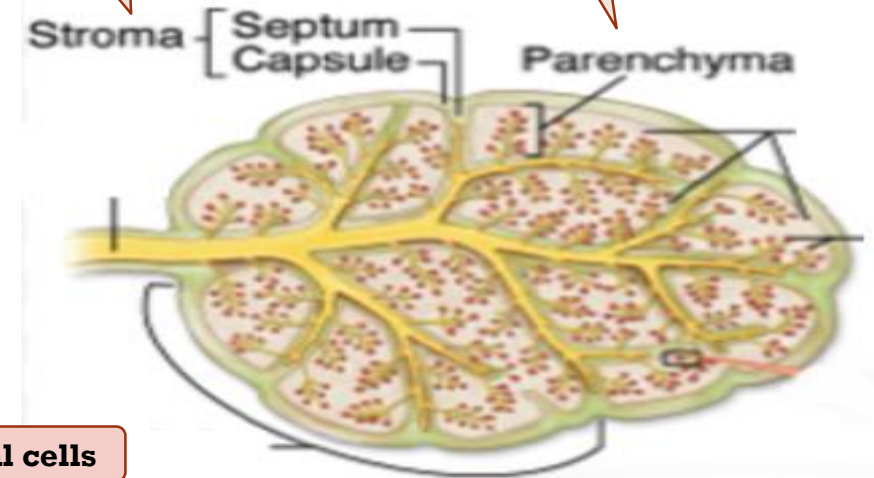
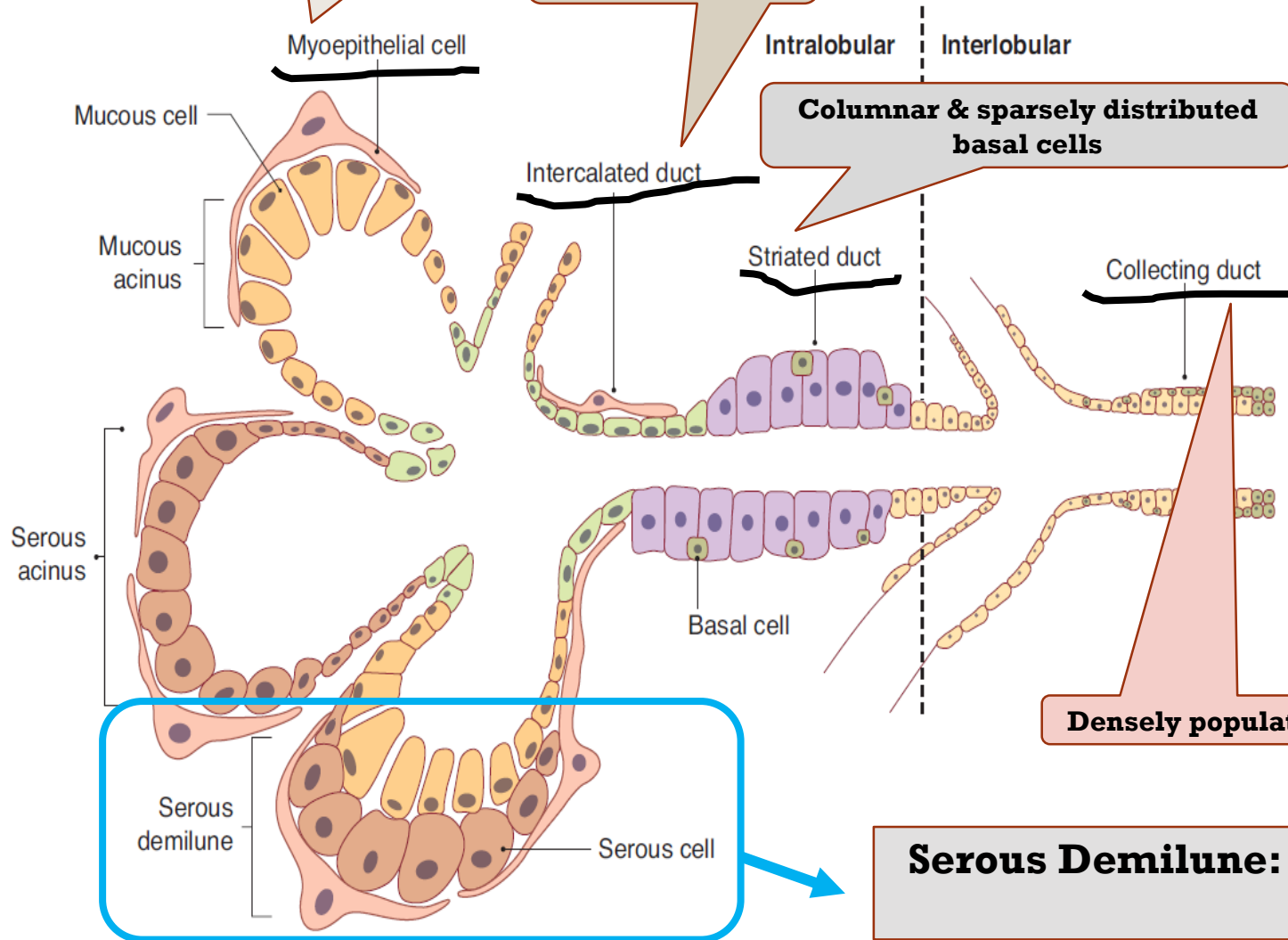
Lined by cuboidal cells

Intralobular Interlobular

Columnar & sparsely distributed basal cells

Supporting connective tissue

Glandular secretory tissue



Serous Demilune: Serous cells form cap or Demilune outside mucous cells.

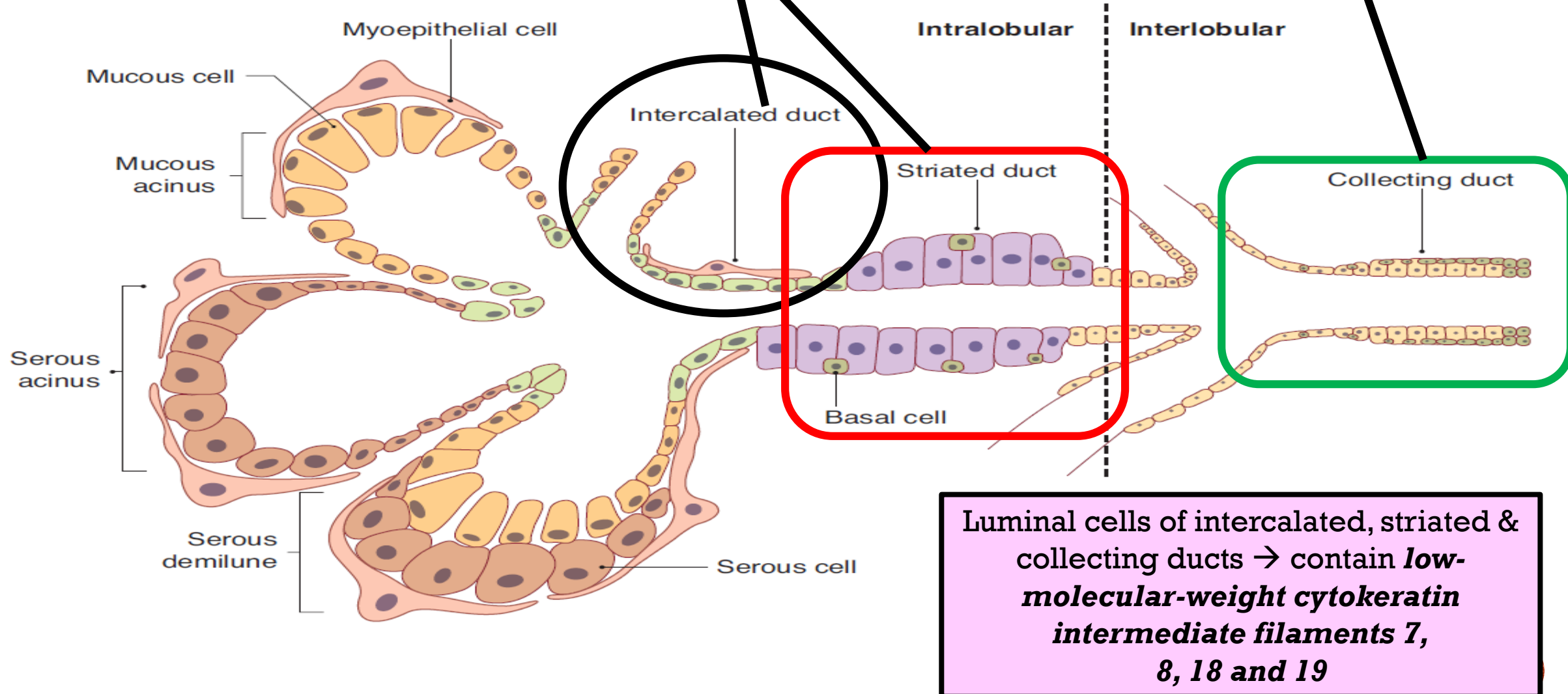


DUCTAL SYSTEM (CONT):

Fig: The secretory and ductal elements in a mixed salivary gland.

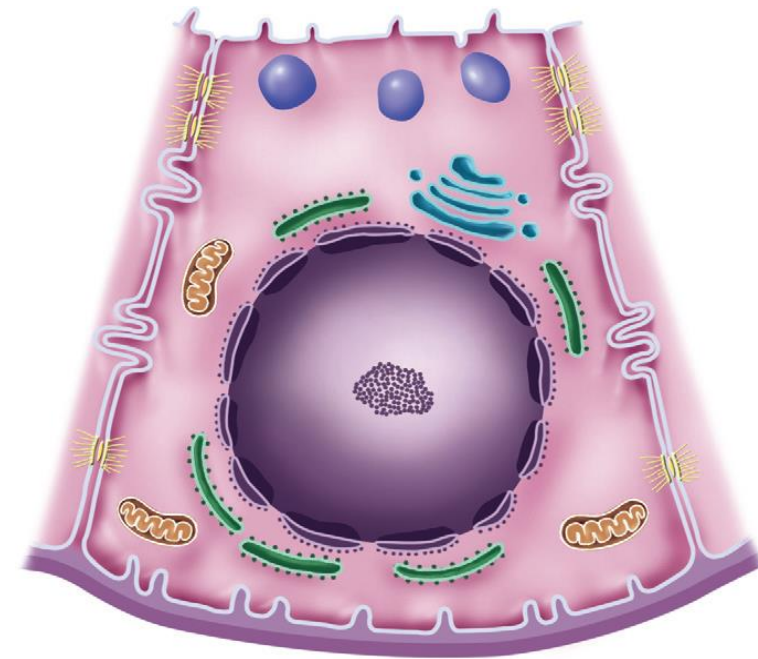
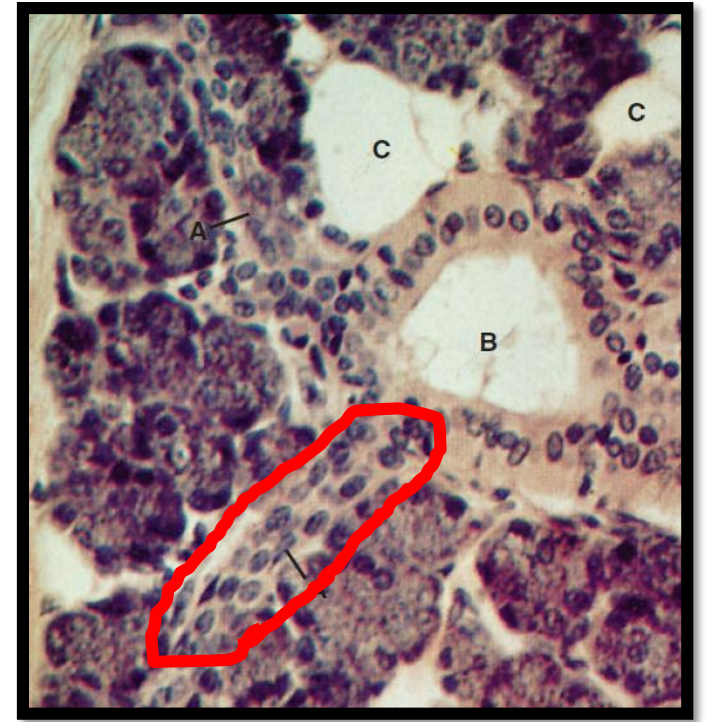
secretory

Salivary transport



INTERCALATED DUCTS:

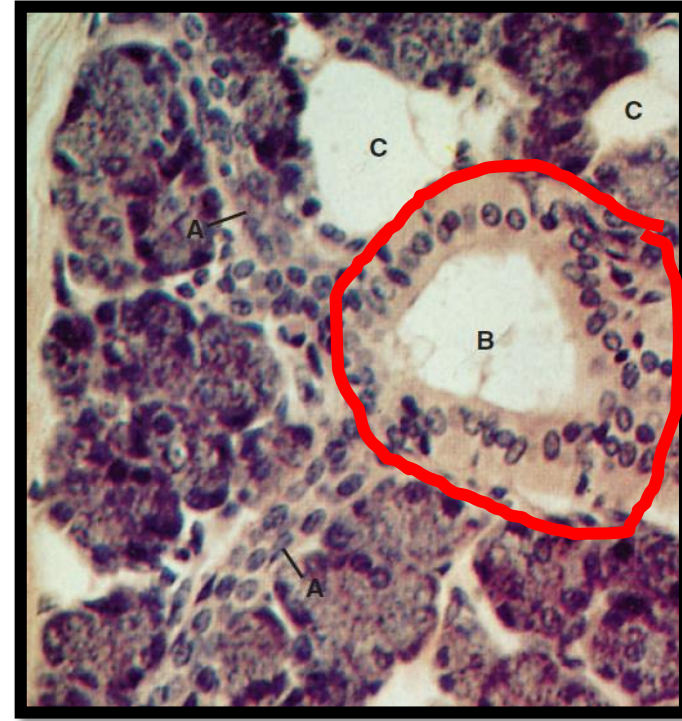
- The smallest (most distal) of ducts.
- Lined by cuboidal epithelial cells.
- Nuclei (prominent) because of relatively scanty cytoplasm.
- Duct cells simple cuboidal tube with smooth luminal & basal surfaces
- Cell to cell junctions – desmosomes
- Apical secretory granules with small amounts of protein synthetic organelles (sometimes)
- Moderate activity of respiratory enzymes, as in the acini and submandibular gland.
- Several acini drain into each intercalated duct.
- In the parotid gland, intercalated ducts are characteristically long, narrow and branching.



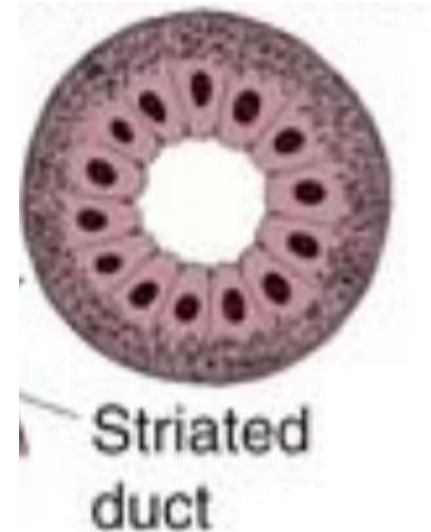
**Intercalated duct cell:
Function → primary
secretion**

STRIATED DUCTS

- Intralobular
- Longer & more active component than intercalated ducts.
- The cells have large amount of cytoplasm
- Large, spherical, centrally positioned nucleus.
- highly polarized cells.
- **Luminal surfaces** - short microvilli.
- **basal (abluminal) surface** - numerous striations under the light microscope (corresponding to multiple infoldings of the plasma membrane at cell base).

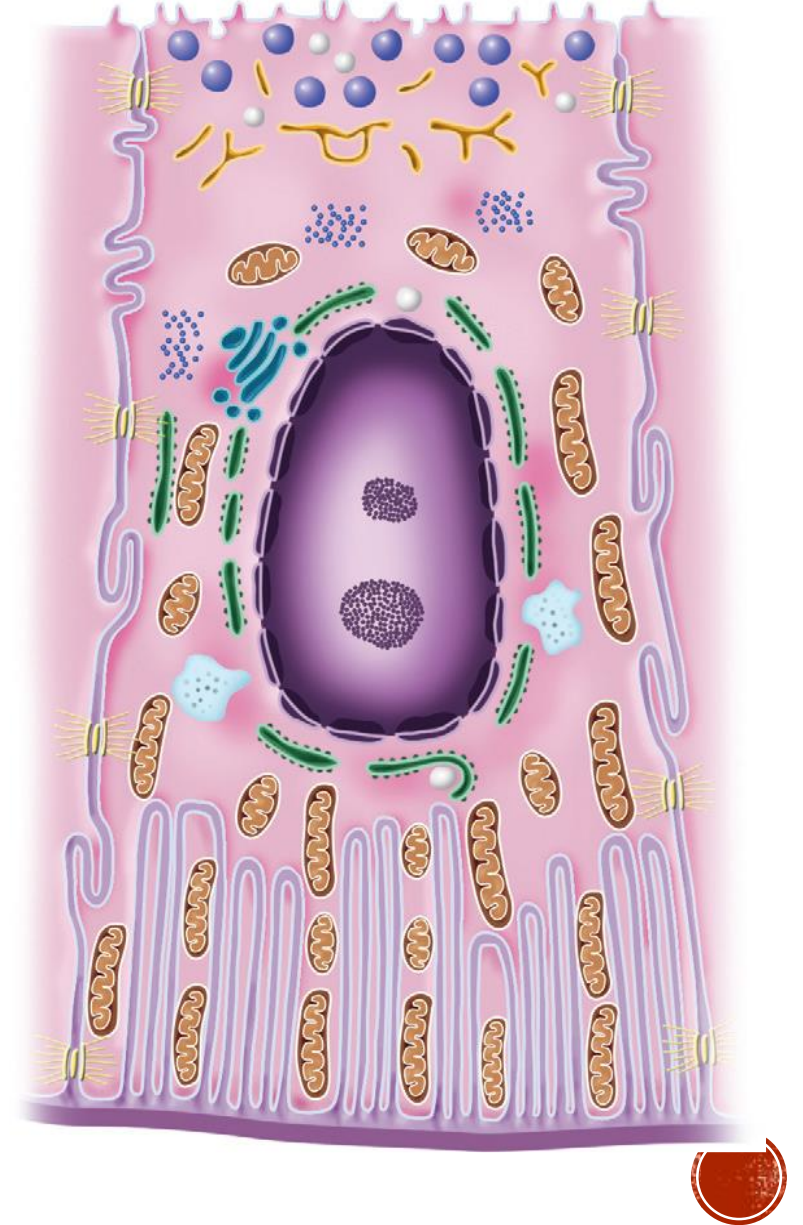


Salivary modification & transport → Conversion of isotonic or slightly hypertonic fluid (concentrations similar to plasma) into hypotonic fluid.



STRIATED DUCTS (CONT.)

- Vertically aligned mitochondria packed between infoldings.
- Adjacent cells intertwined in a complex pattern
- Cells anchored together by desmosomes.
- Large surface area, supplied with $\uparrow$ energy levels from mitochondria $\rightarrow$ clear involvement in active transport.
- The energy is required for electrolyte resorption (Na^+ & Cl^-), against concentration gradient and secretion (HCO_3^-) without loss of water.
- intense activity of respiratory enzymes (in order to effect & maintain concentration gradients & hypotonicity of luminal fluid, as in submandibular gland)
- small secretory granules in luminal region containing epidermal growth factor, lysozyme, kallikrein and secretory IgA (less abundant in humans than other species & less abundant in parotid than in submandibular gland)



COLLECTING DUCTS:

- Interlobular region, 2 cell layers:
 - **Columnar cell layer** (which lacks striations),
 - **Basal cells.**
- Main duct in parotid gland (appearance like many excretory passages) two layers:
 - 1) Mucosa
 - 2) outer connective tissue adventitia.
- Near its termination the lining of the main duct becomes stratified as it merges with the stratified squamous epithelium of the surface oral epithelium (containing keratin intermediate filament types typical of stratified epithelium in the oral mucosa)

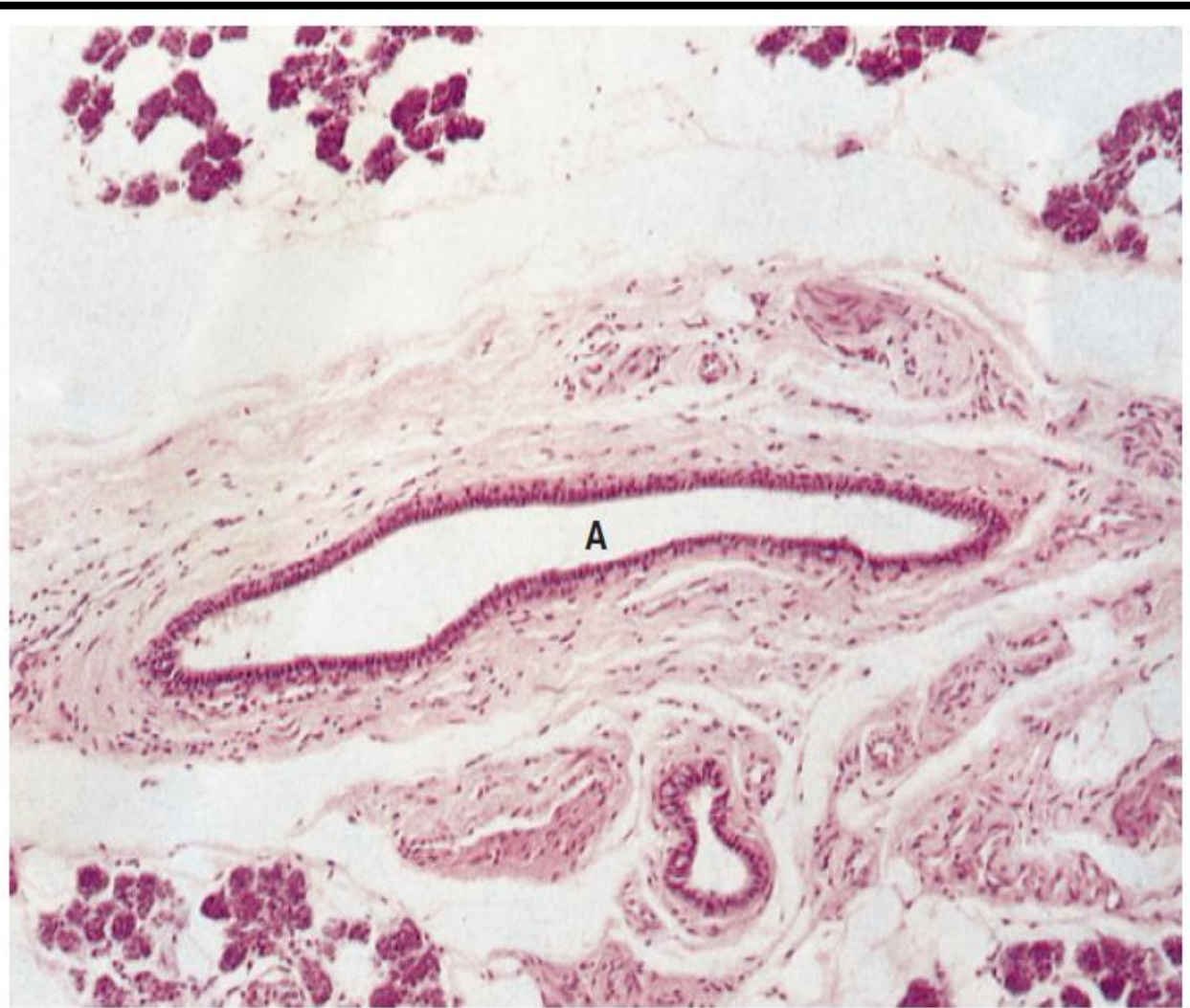


Fig. 16.21 A collecting duct (A) in the parotid gland. As illustrated here, the duct may have two layers of cells (H & E; $\times 60$).

Salivary transport



SALIVARY GLYCOPROTEIN & CALCIUM:

- The salivary glands contain a mixture of salivary glycoproteins that range from neutral to acidic.

Tools for Identification:

- 1) **Alcian blue staining at pH 2.5 → acidic groups (e.g., carboxyl or sulphate groups),**
- 2) **periodic-acid-Schiff → glycol groups**

cell	gland	glycoprotein
serous cells	parotid gland	neutral glycoproteins
mucous cells	submandibular, sublingual & minor salivary glands	acidic glycoproteins
serous cells	submandibular gland	mixture of neutral & acidic glycoproteins
intercalated duct cells		Mixture of both neutral & acidic forms
striated duct cells		neutral glycoproteins



- Ca incorporation into secretory granule → -vely charged glycoprotein parts (ability to repel) → neutralised by +ve charge on Ca.
- molecule condensation in nascent secretory granule.
- The more acidic the glycoprotein, the more Ca (needed).
- Secretory granules of mucous acini & serous acini of submandibular gland → more calcium than the serous acini of parotid gland,
- while secretory granules of striated ducts → no demonstrable Ca
- Release of Ca from secretory granules → precipitation on exposed phospholipids of damaged membranes → sialomicroliths

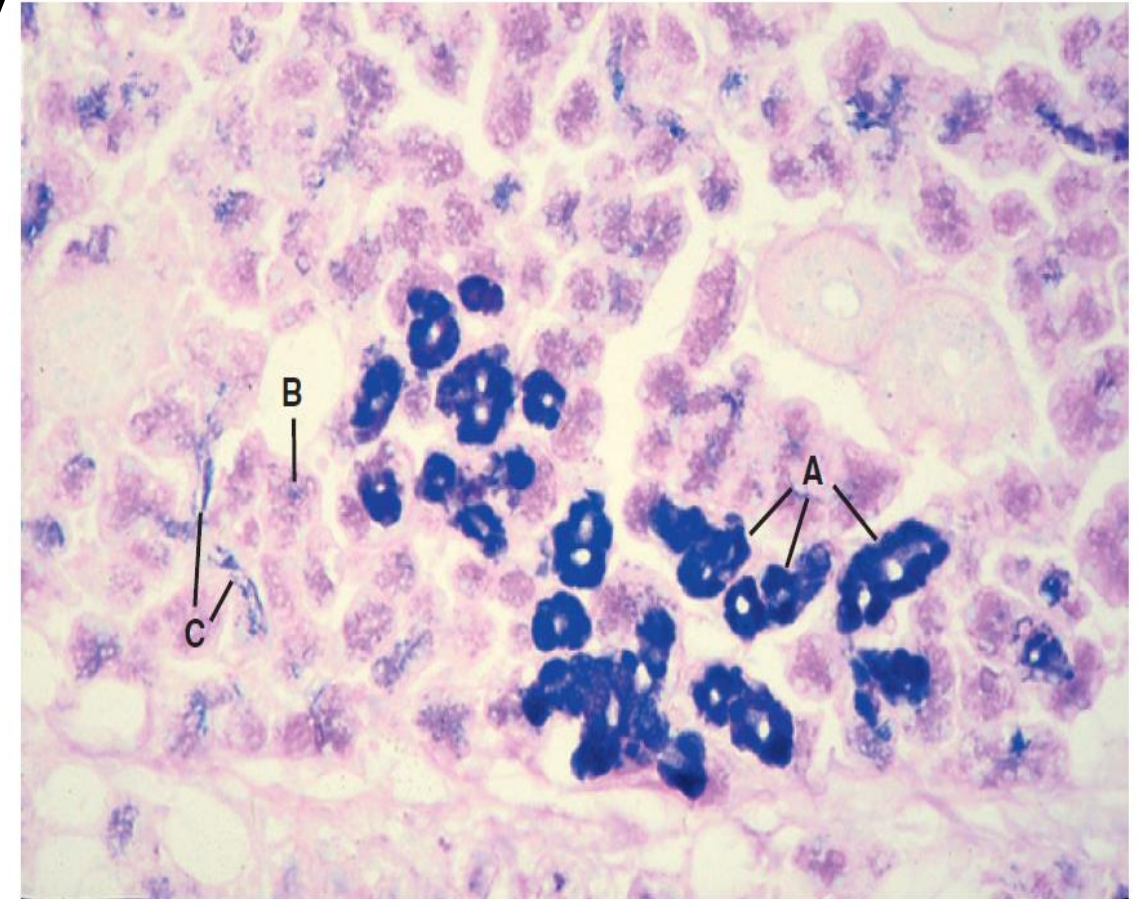
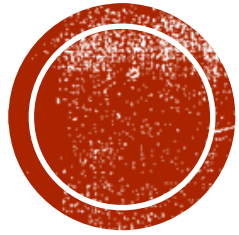


Fig. 16.34 Submandibular gland showing diffuse royal blue staining of acid glycoprotein in mucous cells (A) and discretely stained secretory granules in serous cells (B) and intercalated duct cells (C) that variously contain neutral glycoprotein stained red and acidic glycoprotein stained royal blue (Alcian blue at pH2.5 followed by periodic-acid-Schiff; $\times 190$). Courtesy Dr J D Harrison.

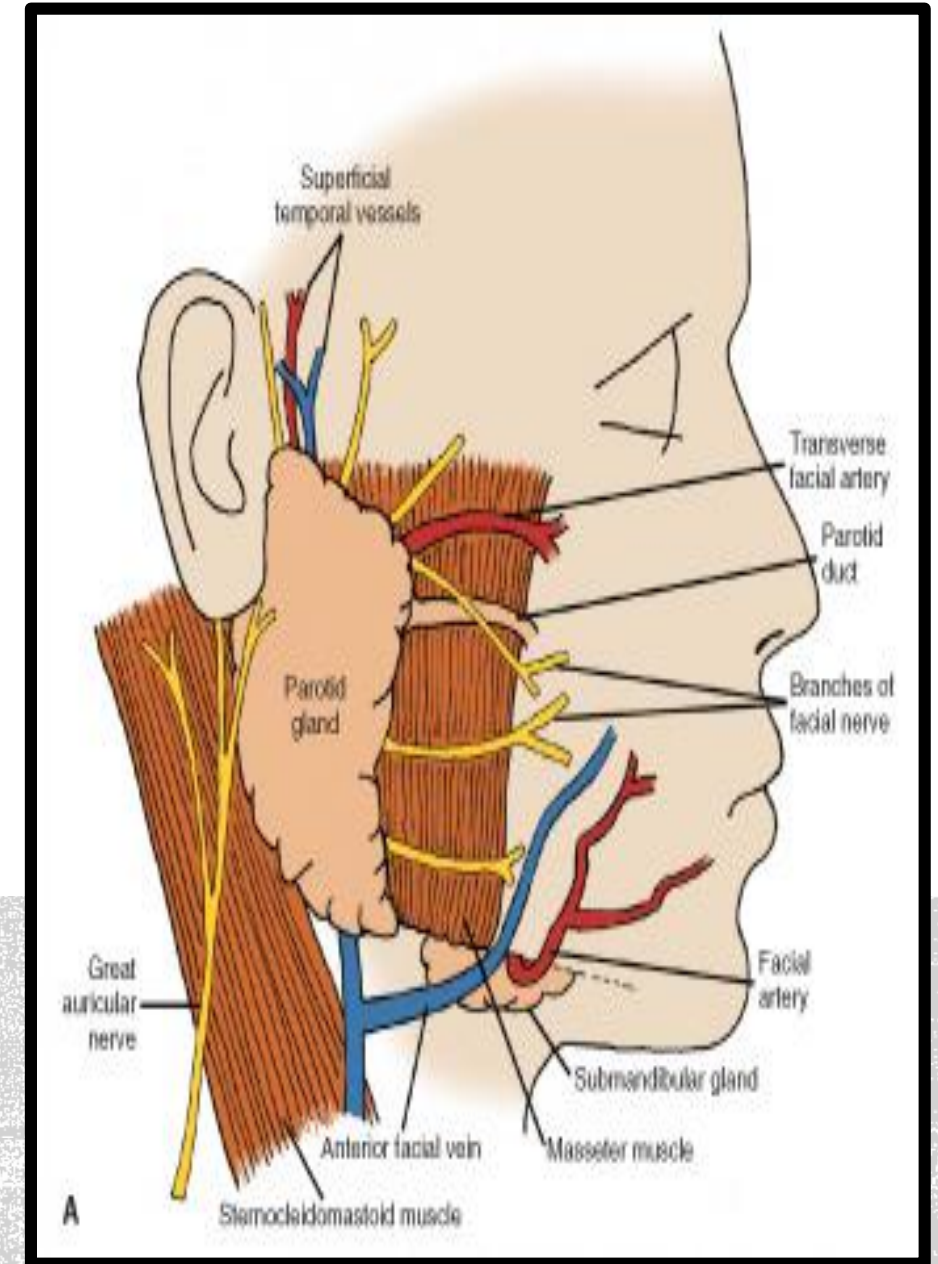
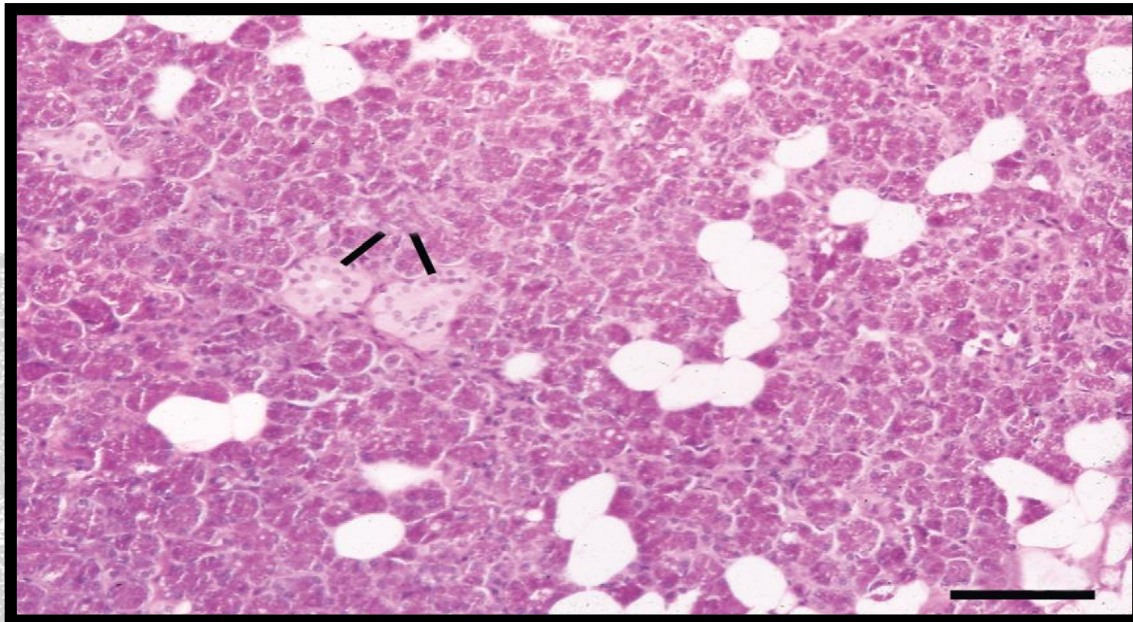


PAROTID GLAND



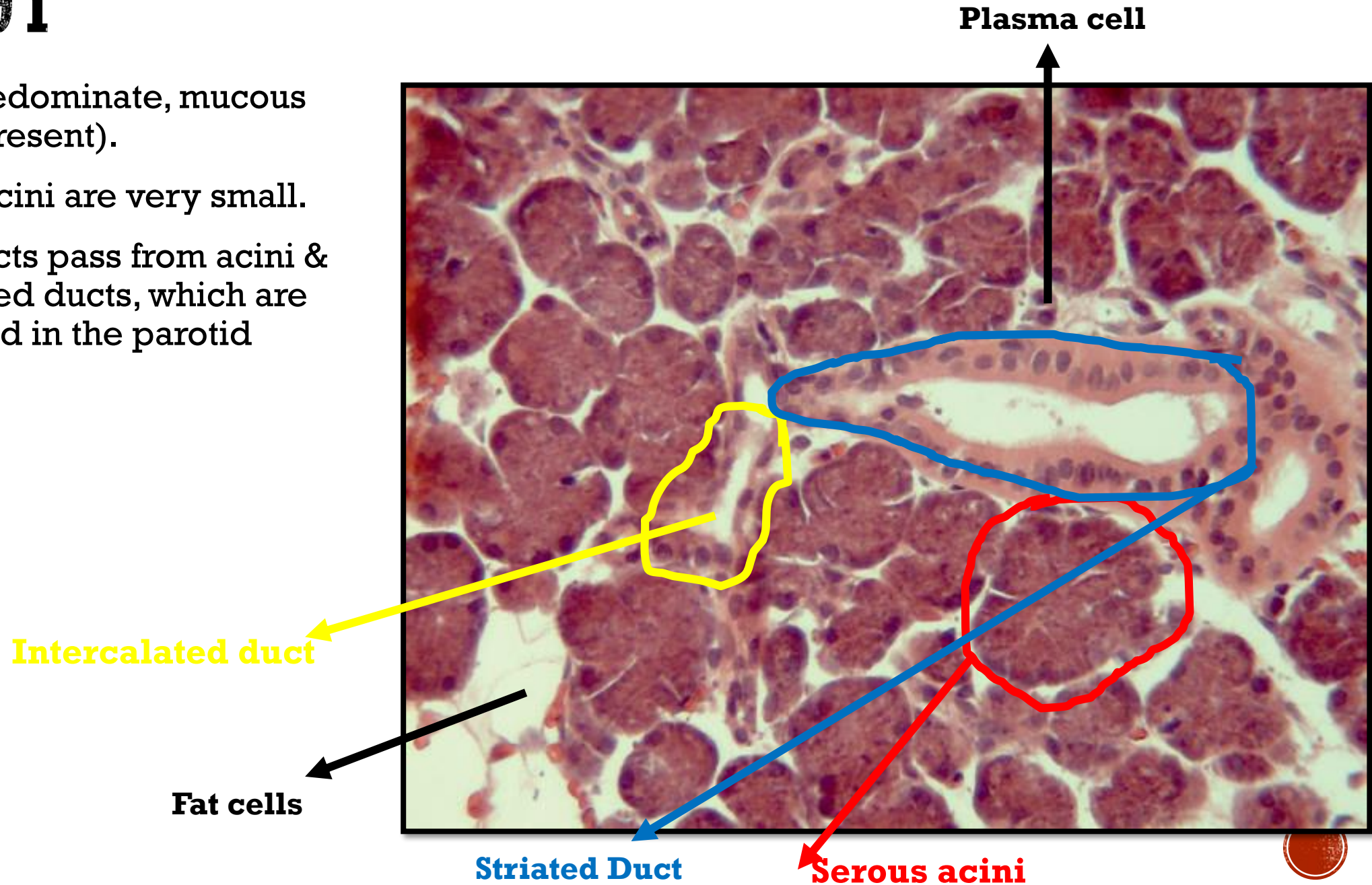
largest salivary gland.

Enclosed within a well-defined capsule, the parotid capsule.



HISTOLOGY

- Serous acini predominate, mucous (occasionally present).
- The lumina of acini are very small.
- Intercalated ducts pass from acini & open into striated ducts, which are well represented in the parotid gland.



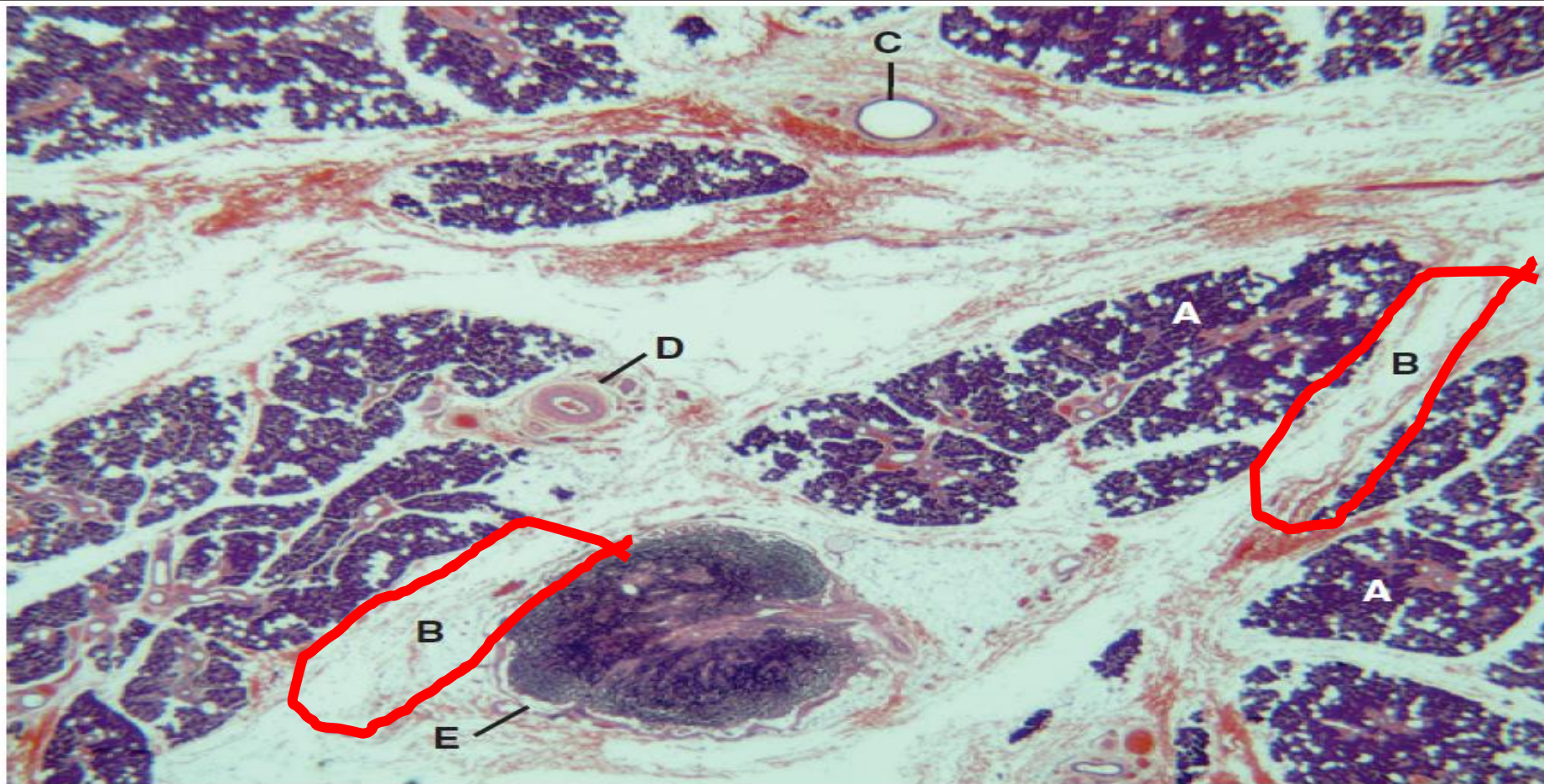


Fig. 16.5 The parotid gland showing the secretory parenchyma (A) being divided into lobules by septa of the stromal connective tissue (B). C = interlobular collecting duct; D = interlobular artery. E = lymph node (H & E; $\times 13$). Courtesy Dr J D Harrison.

- Connective tissue Septa carrying blood vessels nerves divide gland into lobes & lobules.



SEROUS CELLS:

- Characteristic granular appearance with routine H&E staining.
- **Wedge-shaped cell** with broader basal surface surrounding central lumen.
- **Basophilic basal 3rd** : Prominent nuclei are round and located in the **basophilic** (numerous RER present) basal 3rd of the cell, Capillaries are also present in proximity
- **Granular Luminal portion:** Because of numerous refractile (dense, round, secretory) granules in the luminal portion of the cell
- Numerous Infoldings & microvilli of cell membrane.
- Basal lamina separates cell from surrounding CT at basal end of cell.
- Canaliculi run b/w adjacent cells & join lumen.
- Canaliculi & lumen lined by short microvilli.

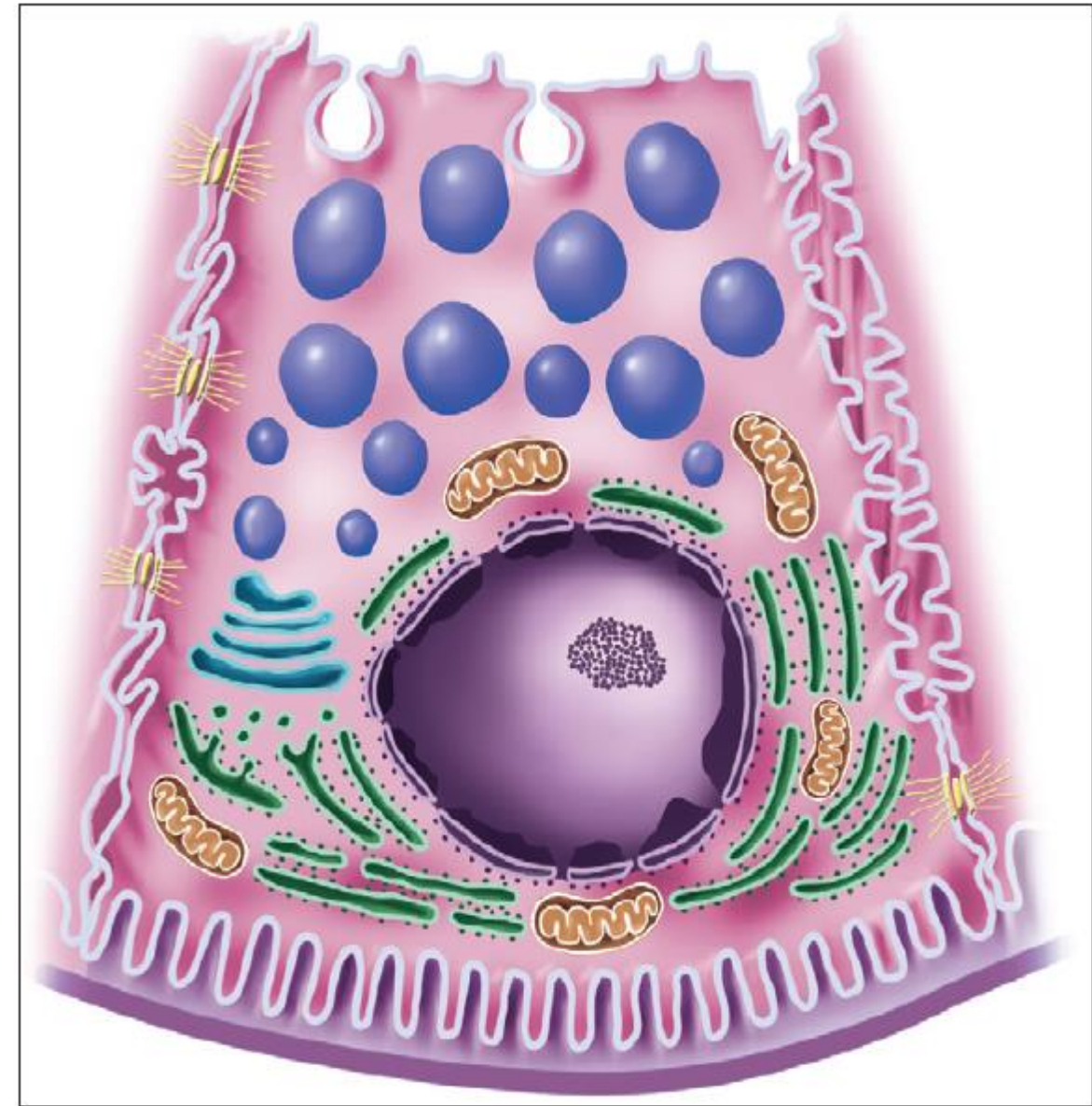
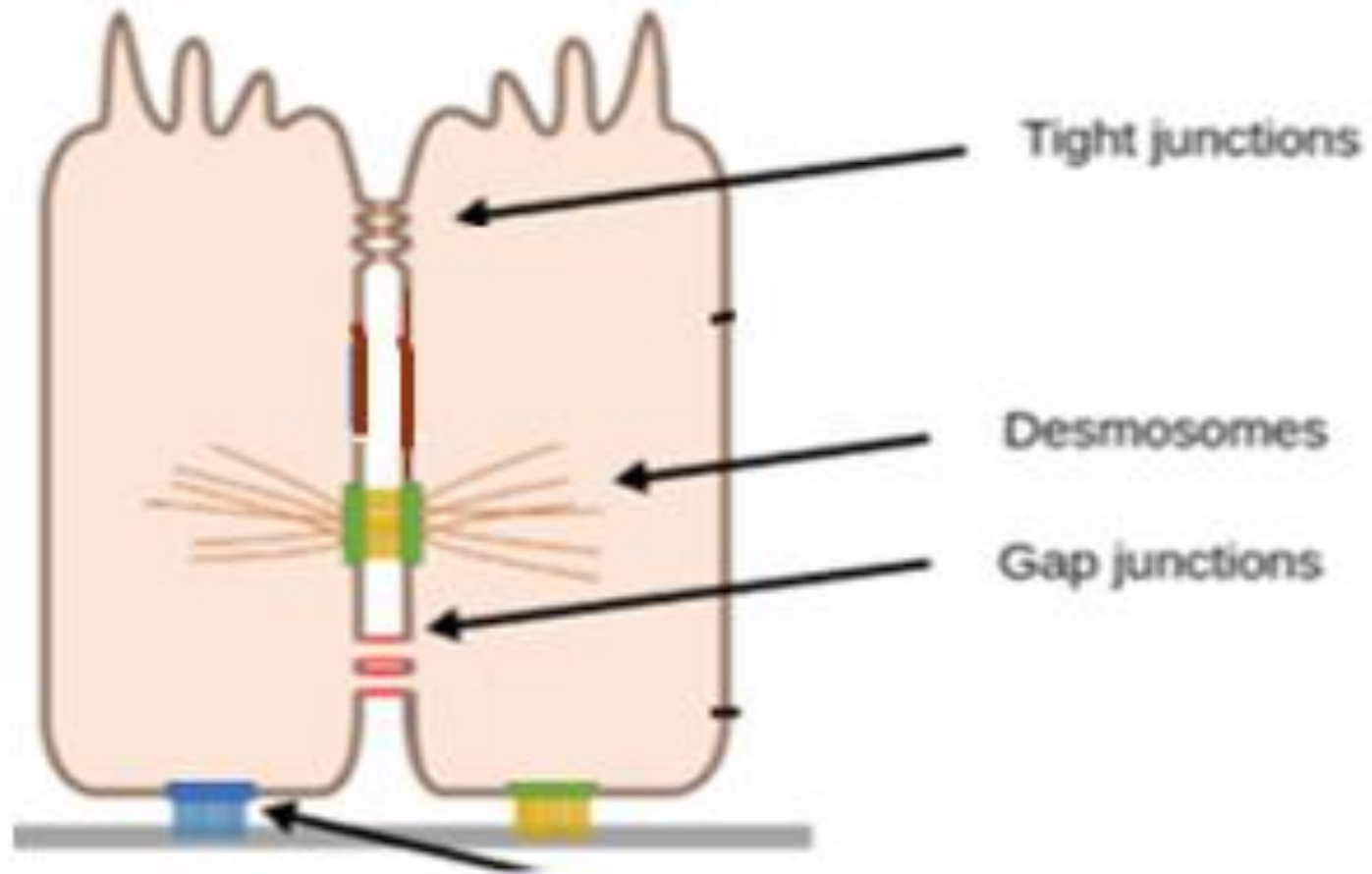


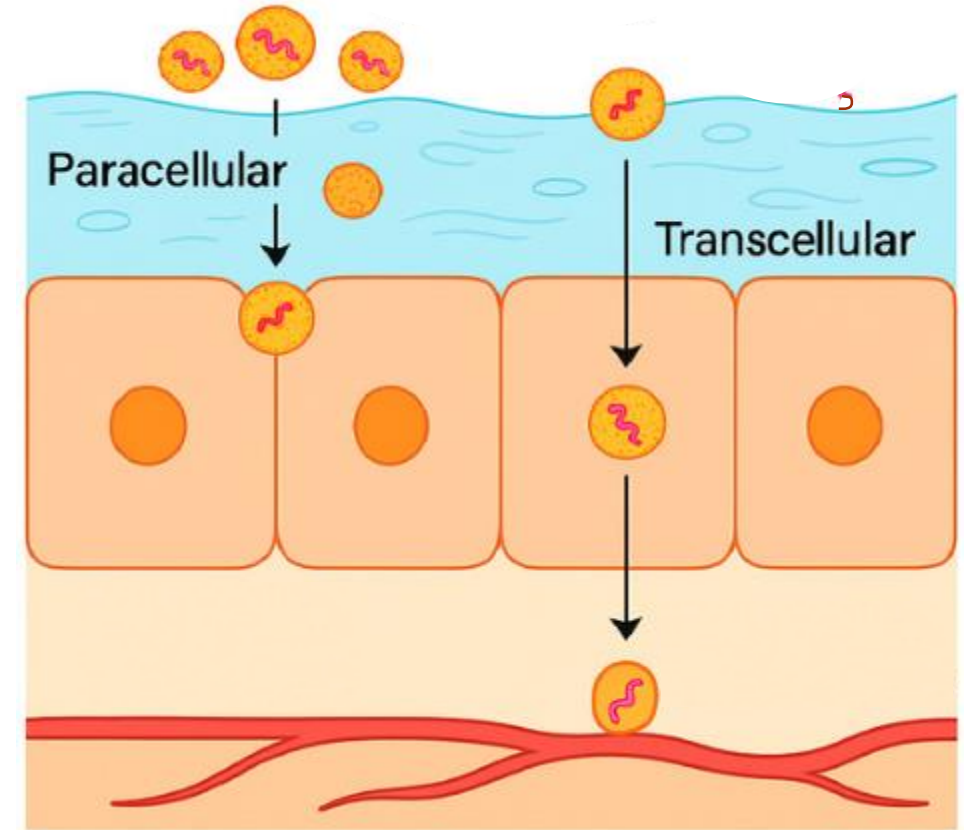
FIGURE 11-11 Serous cell. Intercellular canaliculi are seen in longitudinal (*right*) and cross section (*left*).

Adjacent cell membranes contact at ***desmosomes, gap junctions & tight junctions.***

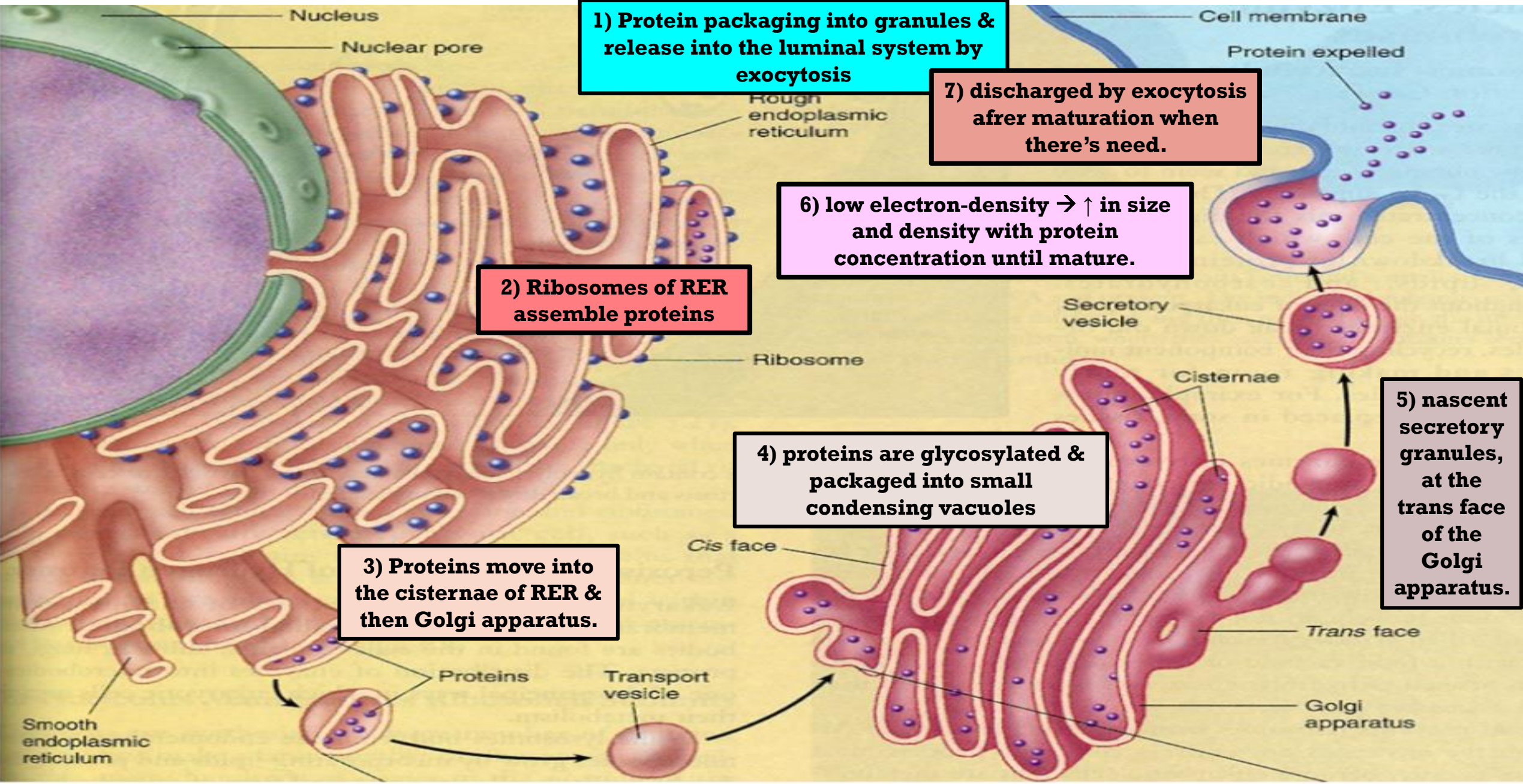


WATER TRANSPORT ROUTES IN SALIVA AND POLARITY OF SEROUS CELLS:

- 99% saliva is water, which passes through 2 routes , as a result of chloride secretion into the lumen.
- **Transcellular Route** : across the cell membrane
- **Paracellular Route**: between adjacent cells
- The serous cells are highly polarized because, the proteinaceous components are packaged in granules for release into the luminal system by exocytosis at the luminal surface of the cells



PROTEIN ASSEMBLY & EXOCYTOSIS IN SEROUS CELLS:



- Mitochondria within the cell supply the energy for:
 - the synthetic process
 - secretory process
- moderate activity of respiratory enzymes can be demonstrated in serous cells, as also in the submandibular gland

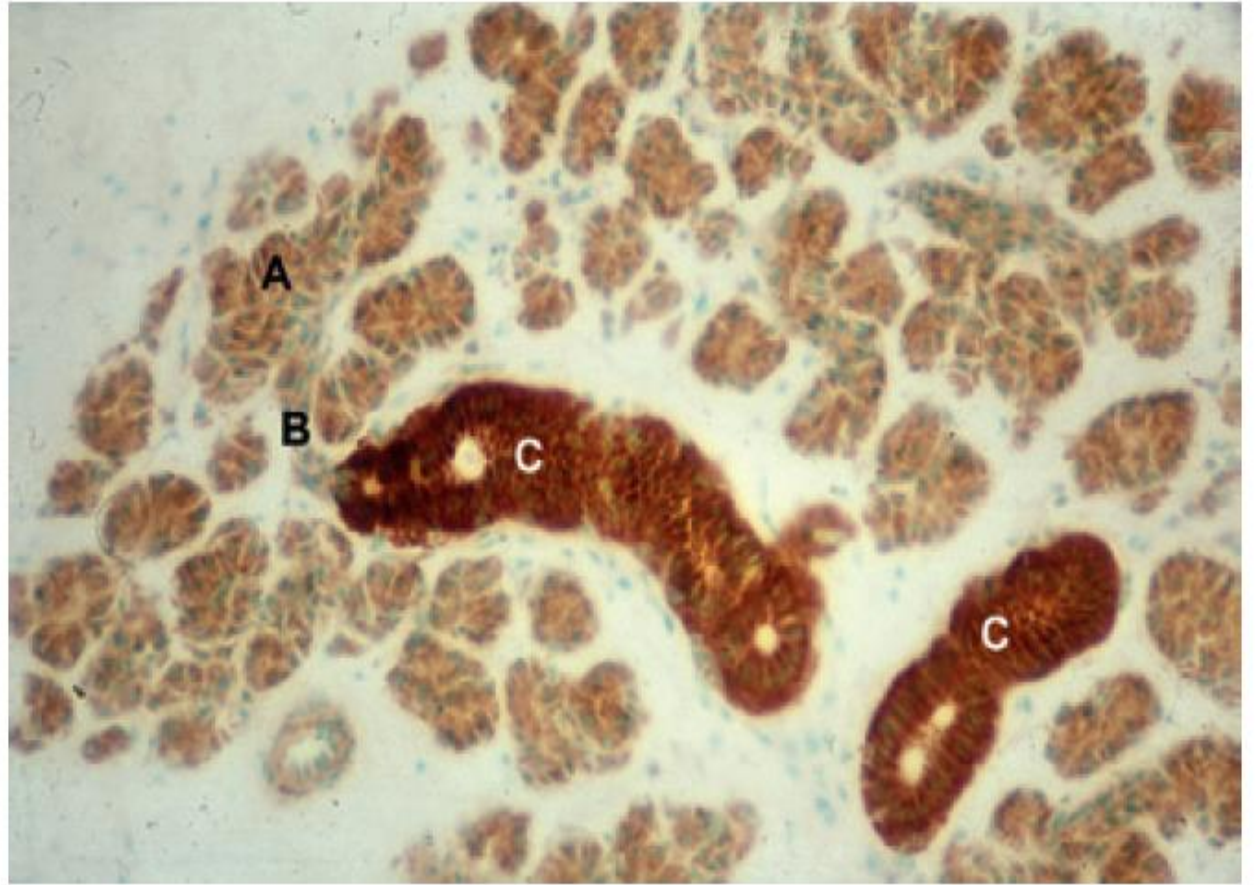
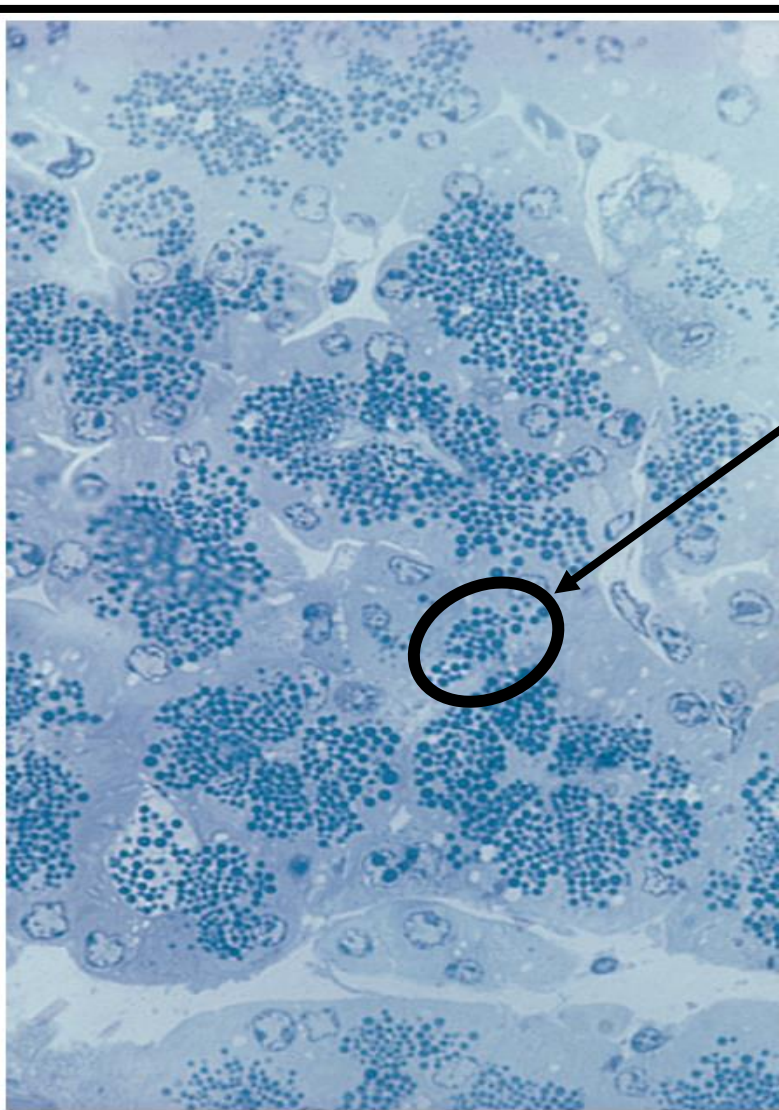
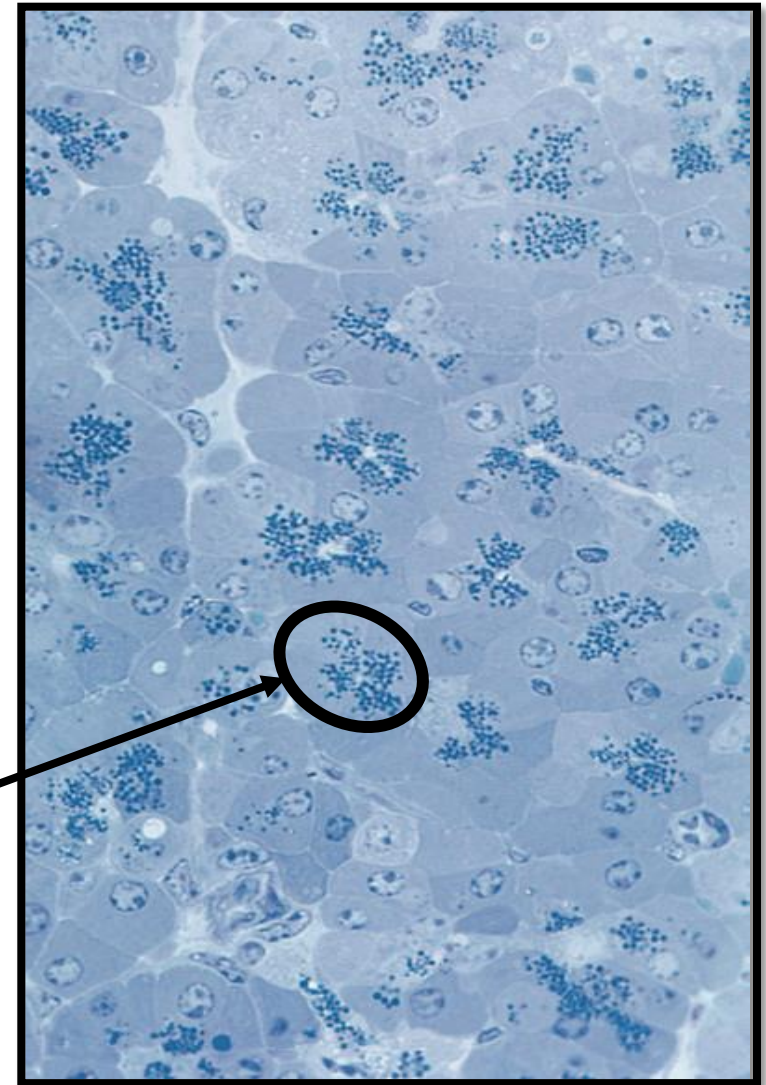


Fig. 16.9 Brown stain shows activity of the respiratory enzyme cytochrome oxidase in submandibular gland with moderate activity in serous acini (A) and intercalated ducts (B) and intense activity in striated ducts (C) ($\times 130$). Courtesy Dr J D Harrison.



Unstimulated (Resting) Serous cell:

Numerous secretory granules (dark blue) in the distal (luminal) parts of their cytoplasm

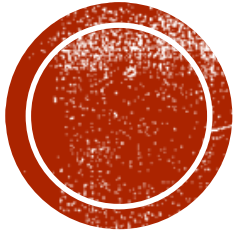


Stimulated Serous cell:

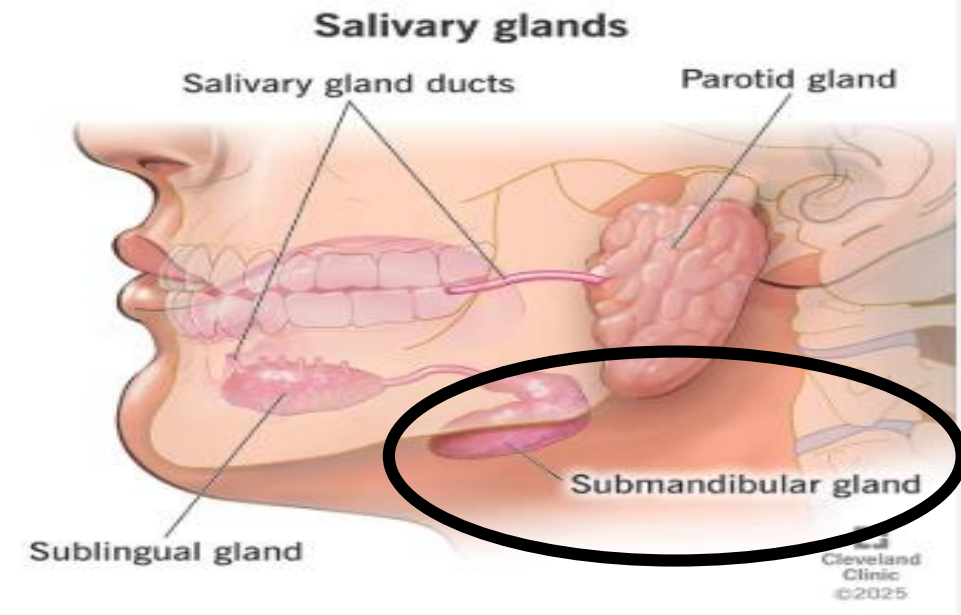
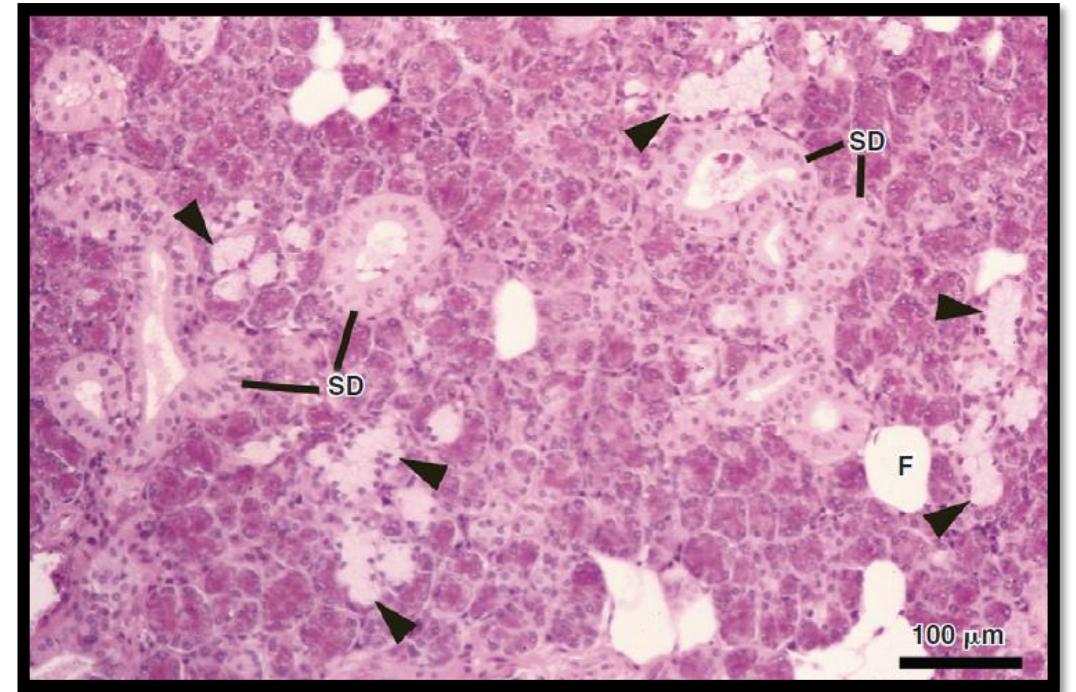
Soon after reflex stimulation (meal times), the secretory granules (dark blue) are depleted after being discharged into the lumen of the acini by exocytosis



SUBMANDIBULAR GLAND

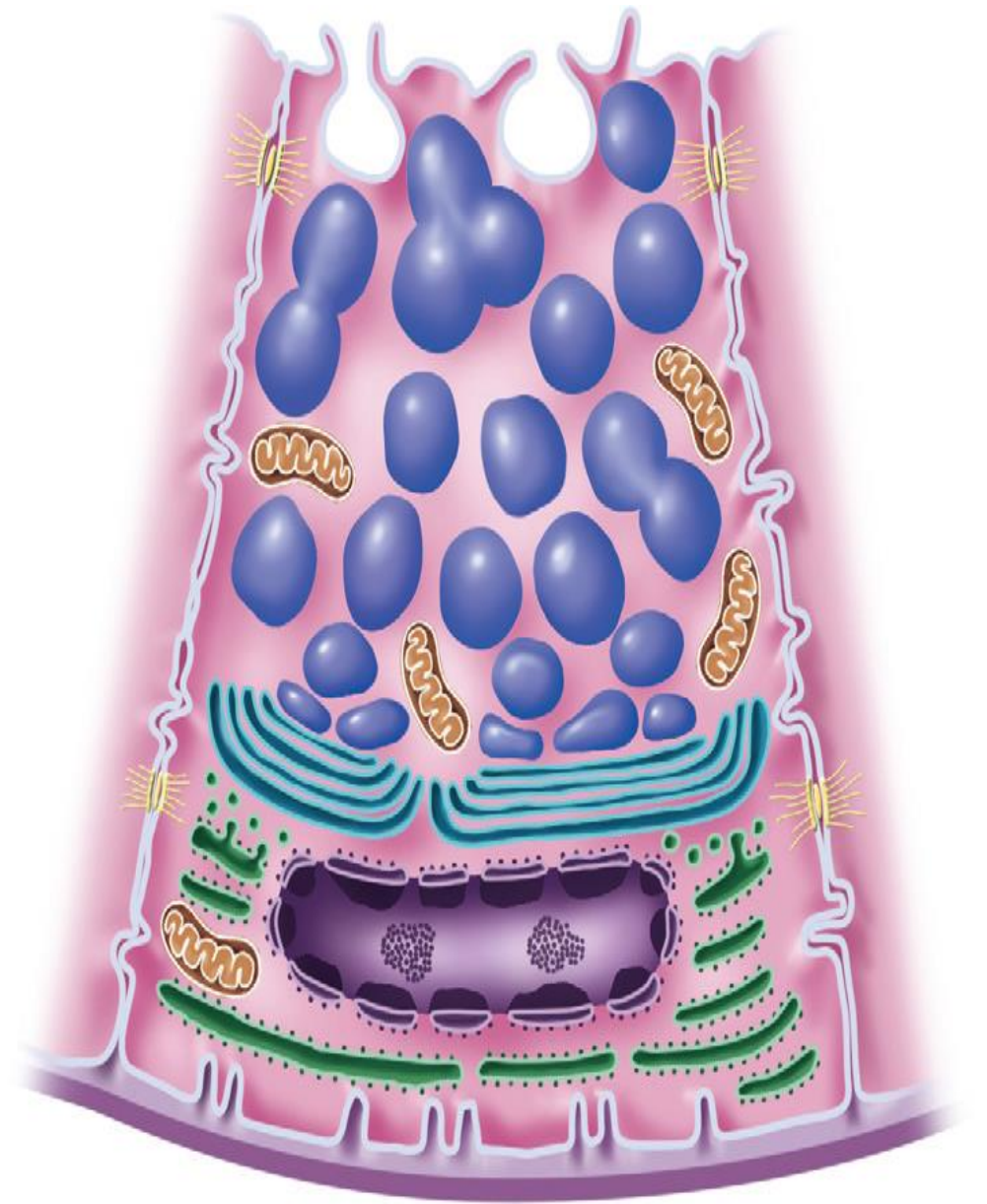


- The second largest of the salivary glands, the submandibular gland,
- produces a mixed serous/ mucous (3:2 ratio) secretion.
- It has a connective tissue capsule.
- The intercalated ducts are shorter and the striated ducts are longer and more conspicuous.



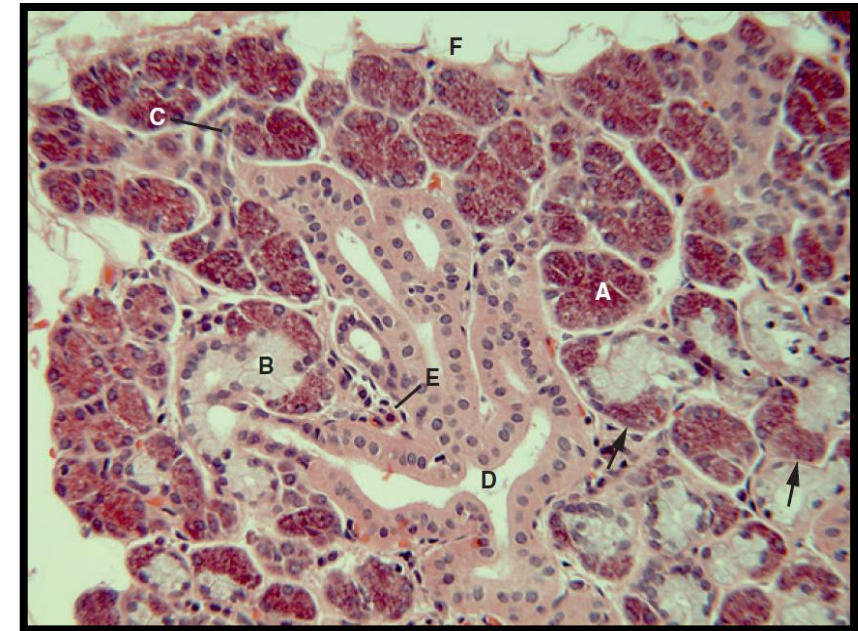
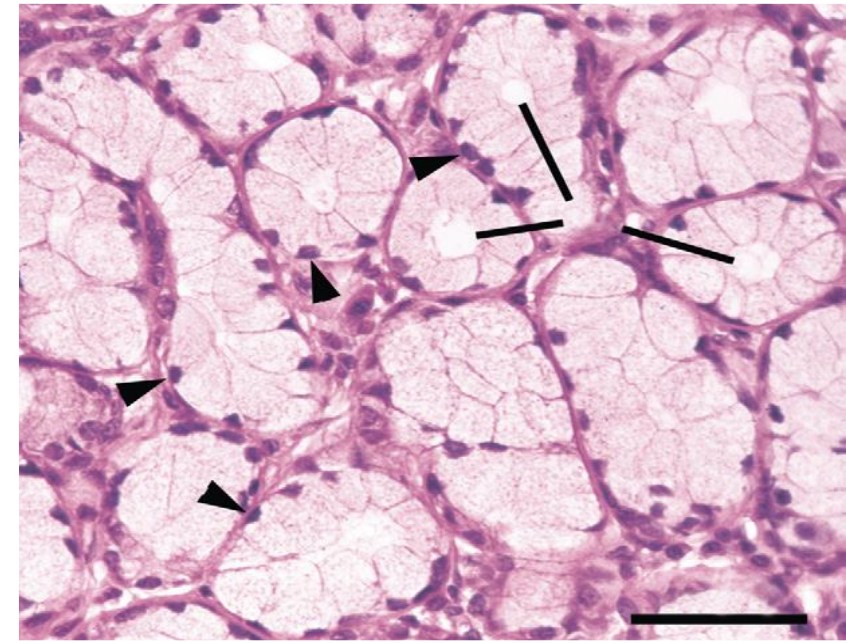
MUCOUS CELLS

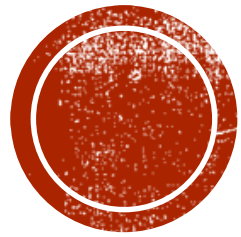
- Mucous acini – paler than serous, since mucin content does not readily take up routine stains.
- nuclei compressed into basal part of cell.
- Small, crescent-shaped collections of serous cells → at most distal ends → referred to as serous demilunes (more recently → artefact of preparation) **changed concept** → serous cells align with mucous cells to surround a common lumen, leaving no demilune structure
- conventional methods fixation → distended mucous cells that displace serous cells towards basal portion of acinus to form demilune structure



MUCOUS CELLS (CONT.)

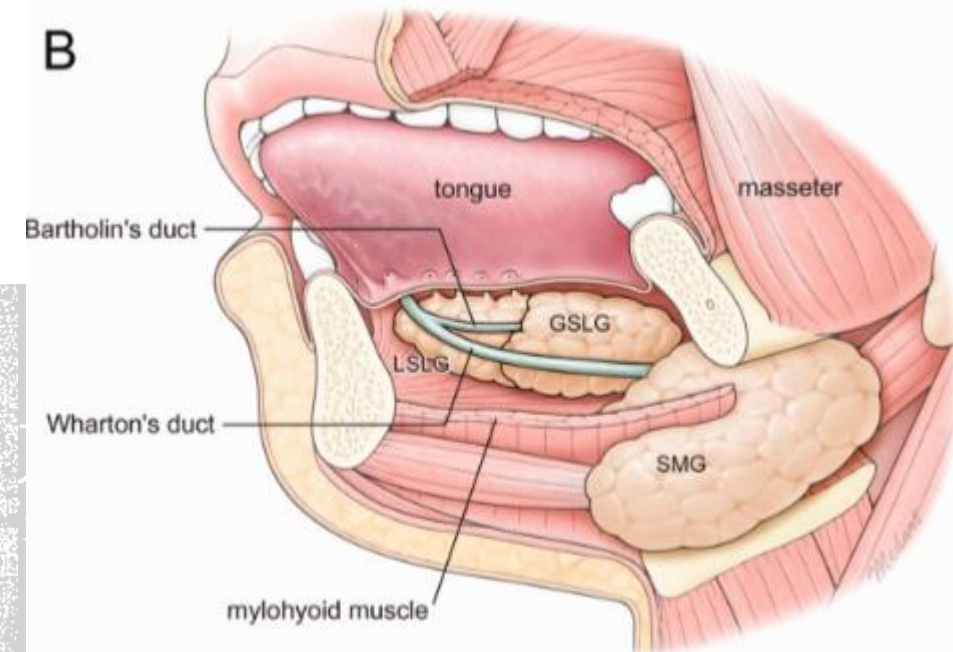
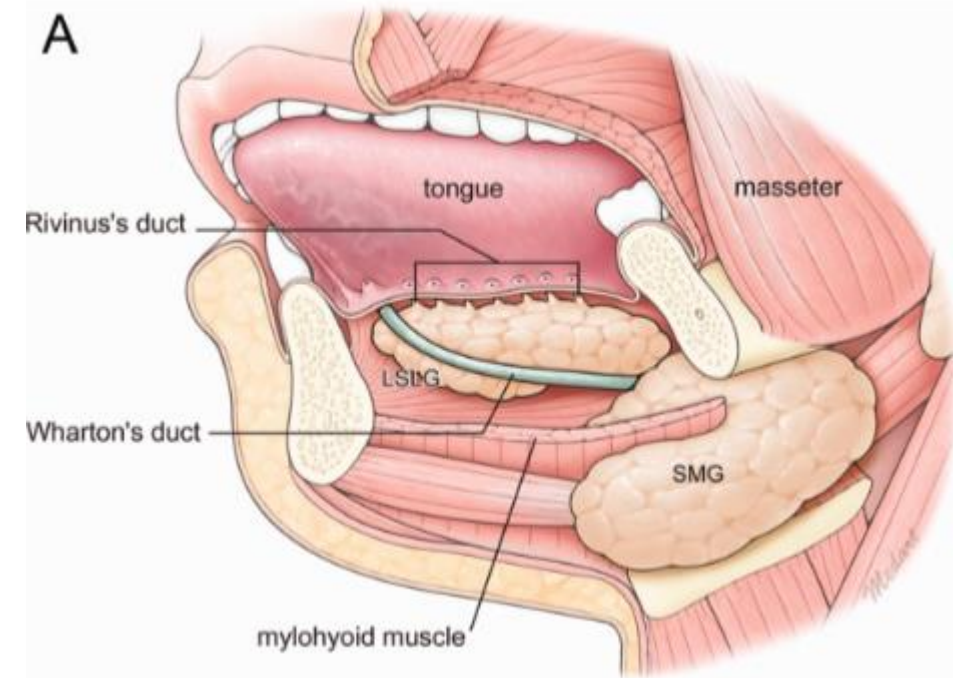
- ***In the early stages of synthesis of secretory products***, large amounts of RER & a Golgi apparatus (conspicuous because of the amount of carbohydrate that is added to the secretory protein) are present in mucous cells, together with a few secretory granules.
- ***At a later phase of its secretory cycle***, the mucous cell exhibits numerous secretory granules (readily distinguished from serous cells by their pale appearance)
- Fewer cell membrane microvilli & infoldings than serous cell.
- Nuclei - round & centrally located (with minimal distortion methods)
- Conventional fixation → distension of mucous secretory granules, → show discontinuous limiting membranes & coalescence to form irregularly-shaped secretory granules, & nucleus (flattened & displaced into basal part of cell)
- Granules discharge into the lumen by exocytosis (same as serous)





SUBLINGUAL GLAND

smallest of the three major pairs of salivary glands
not a single unit, unlike parotid & submandibular glands
Mixed gland (previous opinions) more recent → mucous

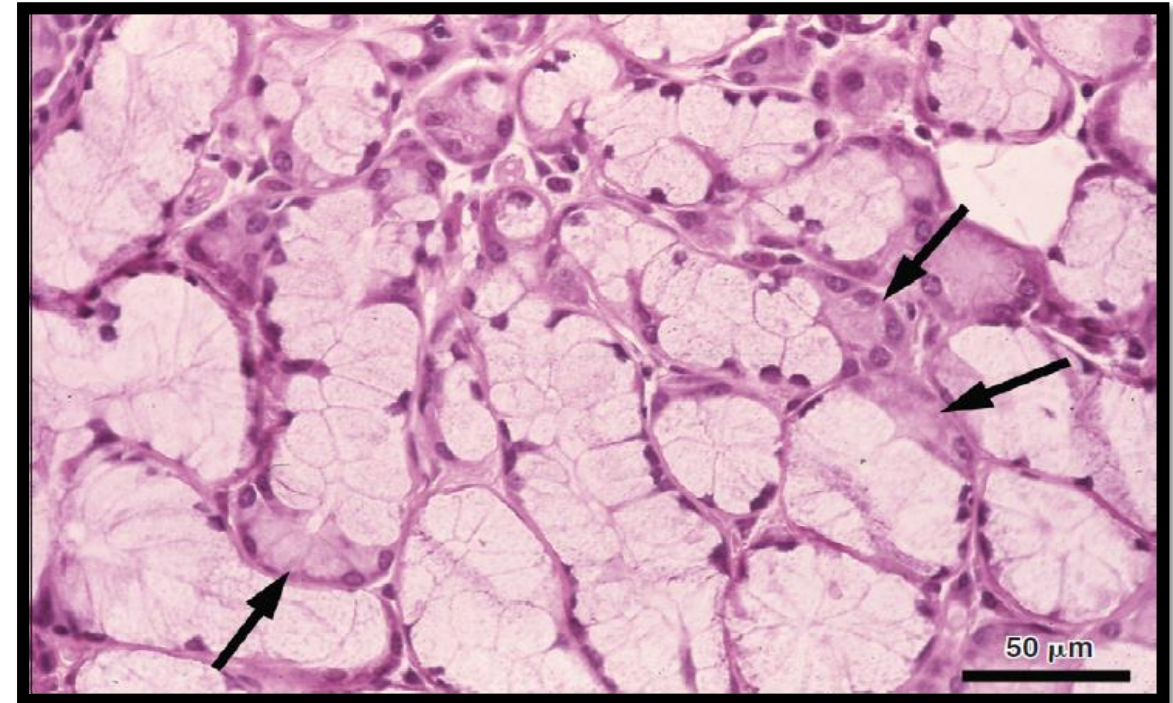


HISTOLOGY

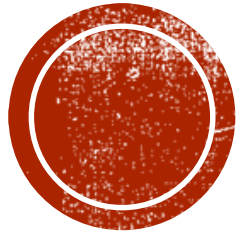
- Consist of many groups of pale staining mucous cells with darker staining so-called serous acini & demilunes (previously)
- Recent research has shown that the so-called serous cells in this gland are immature mucous cells.
- The histological structure of the greater and lesser sublingual glands is identical to that of the mucous minor salivary glands.

DUCT SYSTEM:

- Much less well developed than in the other major salivary glands & striated ducts are usually absent.
- The acini sometimes lead to intercalated ducts but these may be absent & the acini lead to collecting ducts, which lack the basal striations that characterise striated ducts.
- The sublingual saliva is therefore rich in sodium.



MINOR SALIVARY GLANDS



Minor salivary glands have collecting ducts, but intercalated and striated ducts are usually absent.

This duct system therefore does not usually remove salt, so the final saliva released from them is isotonic and rich in sodium.



Table 16.1 Comparison of three major salivary glands

Parotid gland	Submandibular gland	Sublingual gland
Largest	Intermediate	Smallest
Serous	Mixed, mainly serous	Mucous
Intercalated ducts	Intercalated ducts	Few intercalated ducts
Striated ducts present	Striated ducts present	No striated ducts
Collecting ducts end in single main duct	Collecting ducts end in single main duct	Collecting ducts end in many main ducts
Nonspontaneous secretor	Nonspontaneous secretor	Spontaneous secretor
Saliva hypotonic	Saliva hypotonic	Saliva isotonic

SIALOMICROLITHS

- Sialomicroliths are small, hard masses (concrements) that are only evident microscopically in the salivary glands & may occur in the parenchymal cells, in the lumina or in stroma
- Often associated with atrophic foci both increase in number with age.
- Remain small as they are formed in small spaces.
- Different from larger salivary stones (sialoliths)

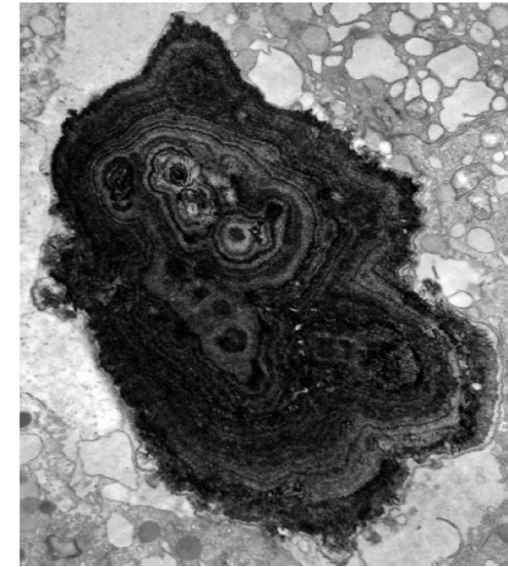


Fig. 16.39 TEM of a sialomicrolith in an acinar lumen. The presence of numerous cores and lamellae indicates that smaller sialomicroliths had accreted (fused together) to form this sialomicrolith ($\times 5,100$). Courtesy Dr J D Harrison.

Contents:

- mineral (in the form of calcium phosphates, including hydroxyapatite)
- nonmineralised components in the form of condensed organic secretory material

Location:

- all normal submandibular glands (related to $\uparrow$ Ca of submandibular gland)
- 20% of normal parotid glands
- rare in sublingual glands and minor salivary glands (b/c of cont. secretion & less stagnation)

FORMATION OF SIALOMICROLITHS

Lysosomes
degrade excess
secretory
granules and
organelles

Autophagosomes
release calcium
during
degradation

Calcium
precipitates on
membrane
remnants

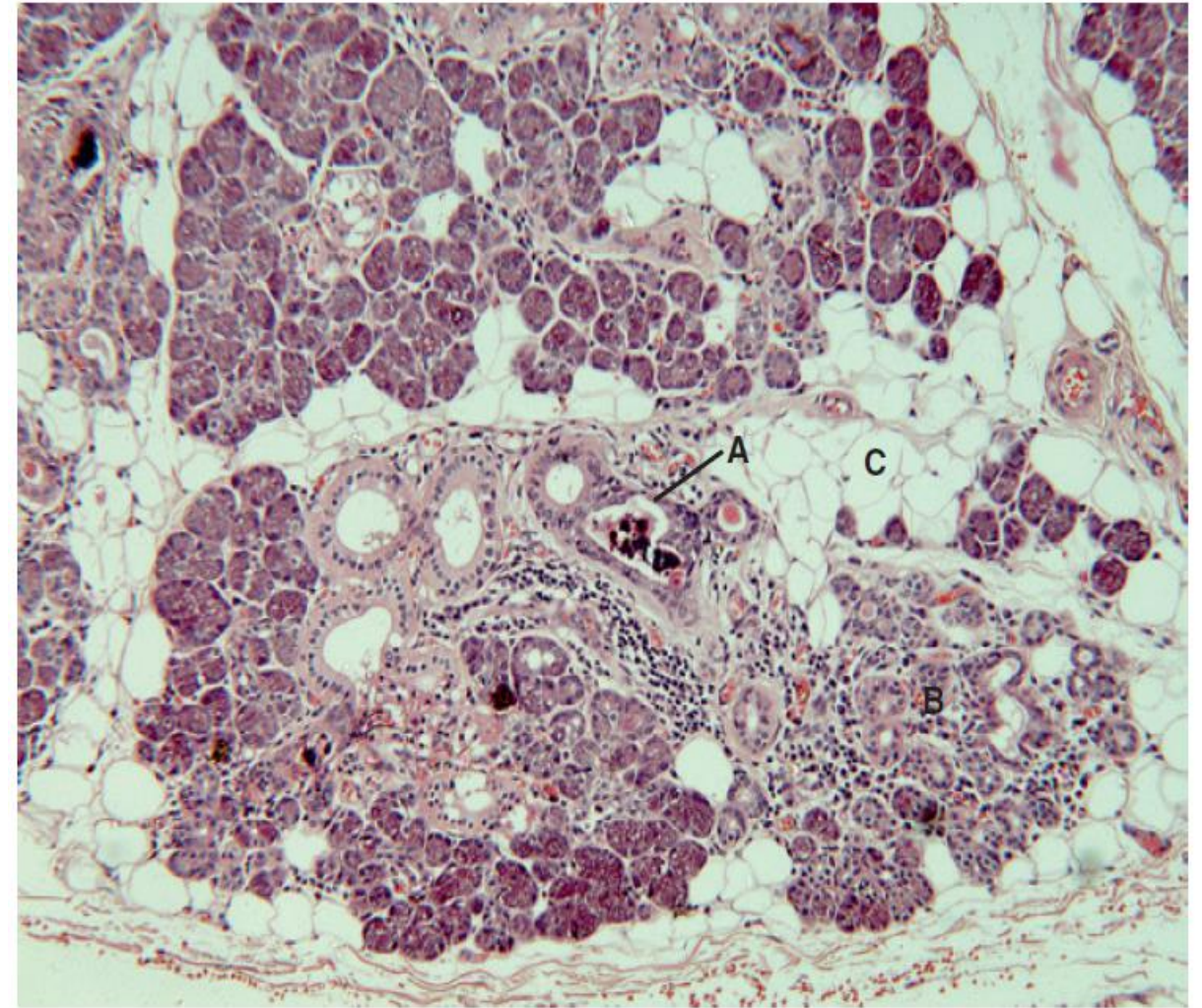


Fig. 16.40 Submandibular gland showing sialomicroliths (A) occluding the lumen of a striated duct that has led to the formation of an inflamed, atrophic focus (B). C = fat cells. Secretory inactivity leads to an increase in sialomicroliths and allows microbes to ascend the ducts and settle in the atrophic foci out of reach of microbicidal saliva (H & E; $\times 95$). Courtesy Dr J D Harrison.

FATE OF SIALOMICROLITHS

Sialomicroliths may form inside cells or lumina

Stagnant secretions favor formation

May pass into duct lumina

Can be flushed out with saliva

May be scavenged by macrophages

Remain microscopic and do not enlarge



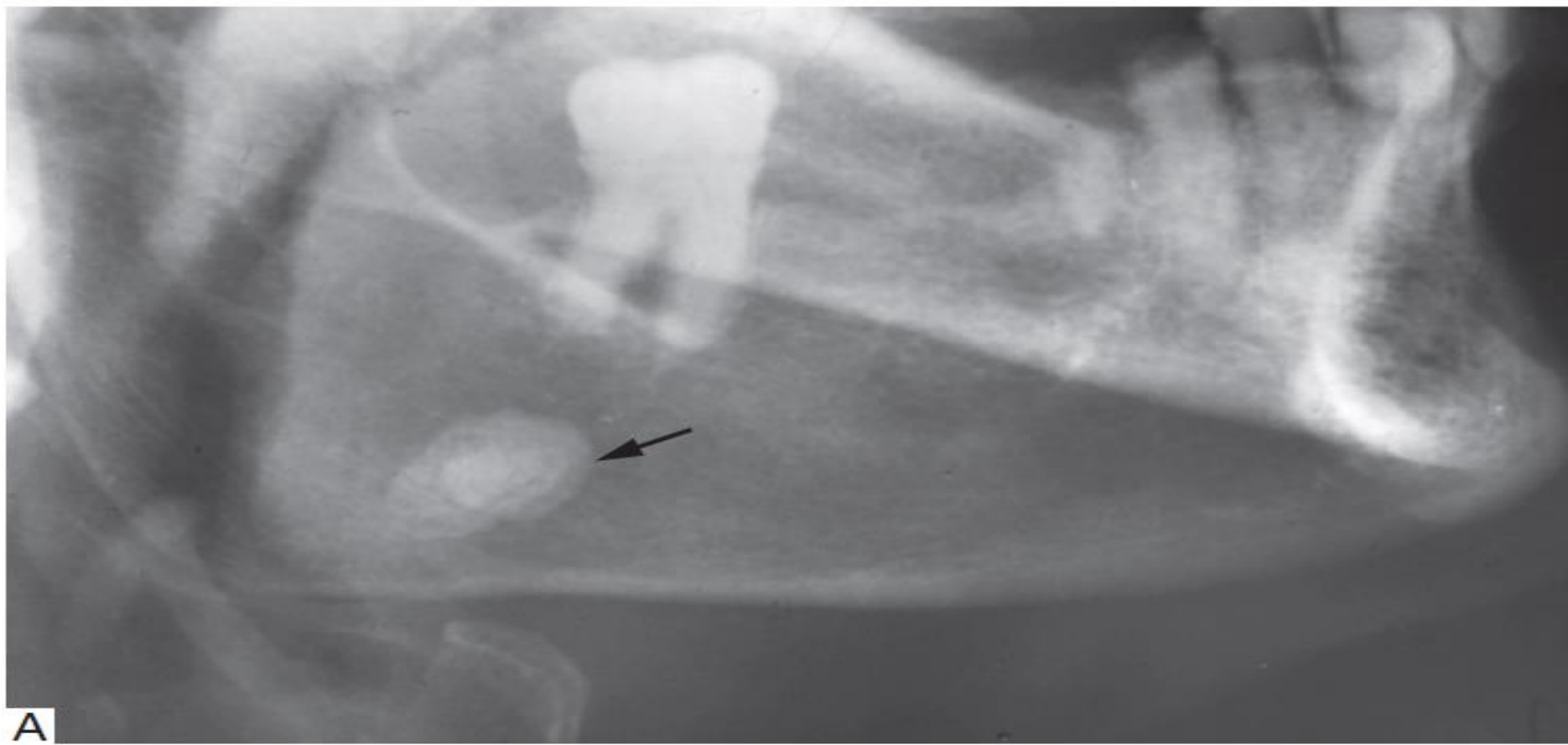


Fig. 16.51 (A) Radiograph showing presence of a sialolith (arrow) at the proximal end of the submandibular duct at the hilum of the submandibular gland. (B) Sialolith removed from a submandibular duct. The length of this sialolith is 2.2 cm. (A) Courtesy Dr J Makdissi. (B) Courtesy Prof. J Freher.



ONCOCYTES

Characteristics :

- Specialized epithelial cells found in acini and ducts
- Appear singly or in small groups
- Strongly eosinophilic on H&E staining
- Granular cytoplasm due to abundant mitochondria
- Commonly increase with age



AGE CHANGES IN SALIVARY GLANDS

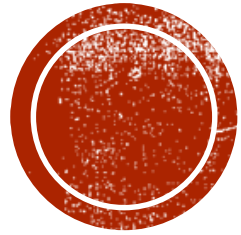
- Decrease in glandular parenchyma after 50 years
- Increase in fibrous tissue and fat cells
- Increase in inflammatory cells and oncocytes
- Duct volume may appear increased
- Acinar shrinkage can mimic duct-like structures



SALIVARY FUNCTION IN AGING

- Aging alone does not necessarily reduce salivary flow
- Healthy elderly individuals usually maintain saliva production
- Xerostomia is not an inevitable result of aging
- Salivary glands possess reserve functional capacity





CLINICAL CONSIDERATIONS



XEROSTOMIA (DRY MOUTH)

- reflection of decreased salivary production associated with the ageing
- Many drugs depress salivary production, (centrally as well as peripherally) with unstimulated salivary flow rates falling from approximately 0.3 mL/min to less than half this value, and some are anticholinergic, such as some antidepressants and antihistamines.
- radiotherapy treatment
- Sjögren's syndrome (where there is invasion and destruction of the parenchyma by lymphoid tissue.)

Remedies:

- Partial relief may be obtained by the frequent administration of artificial salivas.
- Gene therapy

Consequences:
periodontal disease,
dental caries
candidal infections ('thrush').



STAPHNE'S CYST

- A portion of the submandibular gland may invaginate into the lingual surface of the mandible, typically below the mandibular canal, near the angle of the ramus. On a radiograph this will give the appearance of a circumscribed, unilateral radiolucent lesion with a radiopaque border



Fig. 16.45 Radiograph of a Staphne's cavity near the angle of the ramus of the mandible (arrowed). Courtesy Prof. N J D Smith.



MUCOCELES & RANULA

- Damage to the ducts of minor and sublingual salivary glands may result in extravasation of mucus into the surrounding soft tissues.
- When the extravasation persists, an extravasation mucocele is formed.
- In the case of the sublingual gland, it is also termed a ranula because the swelling it causes to the floor of the mouth somewhat resembles the belly of a frog.
- When a ranula is situated above the mylohyoid muscle (oral ranula) it produces a painless swelling that may displace the tongue .
- If it extends below the mylohyoid, either through a mylohyoid hiatus or around the posterior margin of the mylohyoid, it produces a swelling in the neck (cervical or plunging ranula)
- Treatment of these conditions may necessitate the surgical removal of the affected sublingual gland.





Fig. 16.47 Extravasation mucocoele on the lower lip. Courtesy Dr J D Harrison.

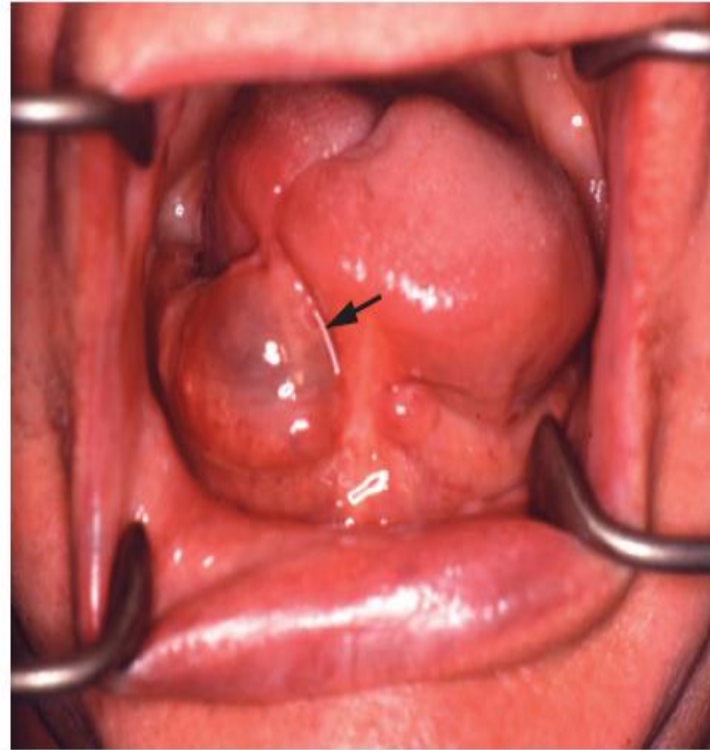


Fig. 16.48 Ranula (arrowed) in floor of mouth caused by trauma that damaged the floor of the mouth and also the tongue. Courtesy Dr J D Harrison.

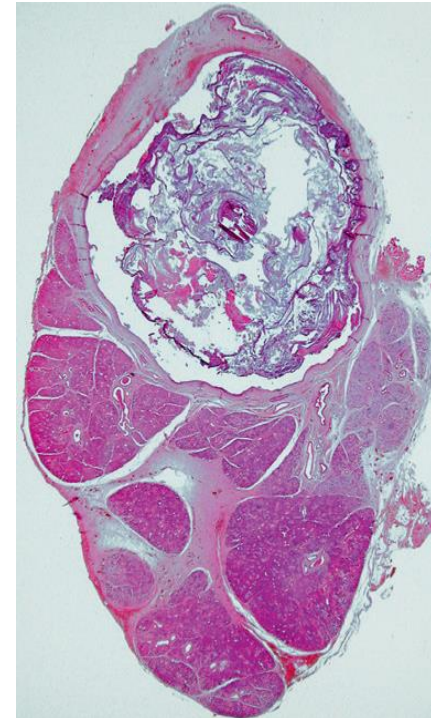


Fig. 16.49 Swelling in the right neck known clinically as a plunging ranula and caused by a damaged sublingual gland from which extravasated saliva is passing through a mylohyoid hiatus. Courtesy Prof. R P Morton and the editor of the *European Archives of Oto-Rhino-Laryngology*. From Samant S, Morton RP, Ahmad Z 2011 Surgery for plunging ranula: the lesson not yet learned? *European Archives of Oto-Rhino-Laryngology* 268: 1513–1518.



SIALOLITH

- Blockage may occur of the main collecting duct of a major salivary gland (usually the submandibular, but occasionally the parotid).
- The cause is usually a sialolith (stone or calculus) but sometimes is a stricture or an inflammatory exudate
- blockage results in swelling of the gland at mealtimes, together with pain and subsequent discharge as saliva gradually flows past the obstruction



SIALOMICROLITHS AND CHRONIC SUBMANDIBULAR SIALADENITIS

- Though present in normal submandibular glands (and less frequently, in normal parotid glands), sialomicroliths may be associated with chronic submandibular sialadenitis (inflammation of the submandibular, which may produce symptoms of pain, swelling and discharge from the gland and sialolithiasis)



Fig. 16.53 Swelling in the neck produced by enlargement of the submandibular gland as a result of chronic obstructive submandibular sialadenitis. Courtesy Dr J D Harrison.





- Mumps is a highly contagious viral infection that primarily targets the **parotid glands**.
- The sensation of pain in mumps caused by enlargement of the gland (with subsequent tension on the unyielding parotid capsule) is mediated by the great auricular nerve.



Sjogren's Syndrome

Dry eyes,
damage to
eye surface



Dry mouth,

Sjögren's (SHOW-gren's) syndrome is a chronic autoimmune disorder where the immune system mistakenly attacks the body's moisture-producing glands, primarily those responsible for making tears and saliva. It is characterized by severe dryness in the eyes and mouth, alongside systemic symptoms like fatigue and joint pain





Thank You

Feeling gratitude and not expressing
it is like wrapping a present and not
giving it.